Deformations of Calabi-Yau varieties

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Abstract

This article gives a systematic introduction to the deformation theory of Calabi–Yau varieties through two complementary approaches: complex geometry and algebra. In the complex-geometric part, we start from the Maurer–Cartan equation, establish the Kodaira–Spencer map, and discuss in detail the role of the Tian–Todorov lemma in solving the equation, deriving the unobstructedness theorem for smooth Calabi-Yau manifolds. In the algebraic part, we construct the Lichtenbaum–Schlessinger T^i functors and the cotangent complex, present the obstruction theory for deformations of schemes, prove Schlessinger's criterion and pro-representability of the deformation functor, and then introduce the Ran–Kawamata T^1 lifting principle together with its application to smooth Calabi-Yau varieties. At the intersection of the two theories, the final part is devoted to threefold Calabi–Yau varieties with ordinary double points. We describe Friedman's smoothability criterion and the subsequent generalization due independently to Kawamata, Ran and Tian: as long as a Kähler resolution exists, deformations are unobstructed, so Friedman's first condition can be dropped.

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1 Deformation theory with approaches of complex geometry

1.1 The Maurer-Cartan equation

A complex manifold is by definition a differentiable manifold M endowed with an atlas whose transition functions are holomorphic. As is known to all, a complex structure defined by such an atlas can also be encoded by an integrable almost complex structure I, which is just an endomorphism of the tangent bundle satisfying a certain non-linear integrability condition. In this section, we will work with the latter description of complex structures. We will explain how to view the various complex structures I on M or, more precisely, the (small) deformations of a given one.

We start out with the set

$$\mathcal{A}_{ac}(M) := \{I \mid I^2 = -\mathrm{id}\} \subset \mathrm{End}(TM)$$

of all almost complex structures on M. First consider an arbitrary $I \in \mathcal{A}_{ac}(M)$. Recall that an almost complex structure I is uniquely determined by a decomposition of the tangent bundle $T_{\mathbb{C}}M = T^{1,0} \oplus T^{0,1}$ with $I = i \cdot \mathrm{id}$ on $T^{1,0}$ and $I = -i \cdot \mathrm{id}$ on $T^{0,1}$. In fact, giving $T^{0,1} \subset T_{\mathbb{C}}M$ is enough, for $T^{1,0} = \overline{T^{0,1}}$.

If $I(t)$ is a continuous family of almost complex structures with $I(0) = I$, one has a continuous family of such decompositions $T_{\mathbb{C}}M = T_t^{1,0} \oplus T_t^{0,1}$ or, equivalently, of subspaces $T_t^{0,1} \subset T_{\mathbb{C}}M$.

Thus, for small t the deformation $I(t)$ of I can be encoded by a map

$$\phi(t) : T^{0,1} \to T^{1,0} \text{ with } v + \phi(t)(v) \in T_t^{0,1}.$$

We write $T^{1,0}$ and $T^{0,1}$ for the subbundles defined by I.

Conversely, if $\phi(t)$ is given, then one defines for small t

$$T_t^{0,1} := (\mathrm{id} + \phi(t))(T^{0,1}).$$

The condition t be 'small' has to be imposed in order to ensure that with this definition $T_t^{0,1}$ has the right dimension. More precisely, t small means that $T_t^{0,1} \subset T_{\mathbb{C}} \to T^{0,1}$ is an isomorphism, which is certainly an open condition, at least on a compact manifold. To avoid problems caused by non-compactness we will henceforth assume that M is compact.

Let us now consider the power series expansion

$$\phi(t) = \phi_0 + \phi_1 t + \phi_2 t^2 + \cdots$$

of a given ϕ. Then $\phi_0 = 0$ and the higher order coefficients $\phi_{i>0}$ will be regarded as sections of $\bigwedge^{0,1} \otimes T^{1,0}$.

Next, we shall study the integrability of the deformed almost complex structures $I(t)$. From now on we will assume that I is integrable and we will denote the complex manifold (M, I) by X. In particular, the higher order coefficients ϕ_i correspond to elements in $\mathcal{A}^{0,1}(\mathcal{T}_X)$.

The integrability condition for $I(t)$ can be expressed as

$$[T_t^{0,1}, T_t^{0,1}] \subset T_t^{0,1}.$$

Let us rephrase this in terms of $\phi(t)$. First note that there exists the $\bar\partial$-operator on the holomorphic tangent bundle $\mathcal{T}_X$ of $X=(M,I)$, which can in particular be applied to $\phi(t)\in\mathcal{A}^{0,1}(\mathcal{T}_X)$. Secondly, one defines a 'Lie bracket'

$$[\ ,\ ]:\mathcal{A}^{0,p}(\mathcal{T}_X)\times\mathcal{A}^{0,q}(\mathcal{T}_X)\longrightarrow\mathcal{A}^{0,p+q}(\mathcal{T}_X)$$

by taking the Lie bracket in $\mathcal{T}_X$ and the exterior product in the form part $\mathcal{A}^{0,*}$. Spelled out in local coordinates this gives for $\alpha=\sum d\bar z_I\otimes v_I$ and $\beta=\sum d\bar z_J\otimes w_J$ with $v_I=\sum\alpha_{Ij}\frac{\partial}{\partial z_j}$ and $w_J=\sum\beta_{Jl}\frac{\partial}{\partial z_l}$ the following formula

$$[\alpha,\beta]=\sum(d\bar z_I\wedge d\bar z_J)[v_I,w_J]=\sum_{I,j,J,l}(d\bar z_I\wedge d\bar z_J)\big[\alpha_{Ij}\frac{\partial}{\partial z_j},\beta_{Jl}\frac{\partial}{\partial z_l}\big].$$

This is indeed well-defined, as any holomorphic coordinate change would change the tangent vector fields v_I,w_I by antiholomorphic factors $\bar f$ and $\frac{\partial\bar f}{\partial z_m}=0$.

Remark 1.1.1. Using the local description, one easily checks that $\bar\partial[\alpha,\beta]=[\bar\partial\alpha,\beta]\pm[\alpha,\bar\partial\beta]$. In particular, the Lie bracket of two $\bar\partial$-closed sections in $\mathcal{A}^{0,*}(\mathcal{T}_X)$ is again closed and the bracket of a closed one and an exact one is exact. Thus, the bracket induces a map

$$H^p(X,\mathcal{T}_X)\times H^q(X,\mathcal{T}_X)\longrightarrow H^{p+q}(X,\mathcal{T}_X).$$

Lemma 1.1.2. *The integrability equation $[T_t^{0,1},T_t^{0,1}]\subset T_t^{0,1}$ is equivalent to the* ***Maurer-Cartan equation***

$$\bar\partial\phi(t)+[\phi(t),\phi(t)]=0. \tag{1}$$

Proof. The assertion can be proved locally. We will omit the parameter t in the notation and shall write

$$\phi=\sum_{i,j}\phi_{ij}d\bar z_i\otimes\frac{\partial}{\partial z_j}.$$

If the integrability condition holds true, then we have in particular

$$\left[\frac{\partial}{\partial\bar z_i}+\phi\Big(\frac{\partial}{\partial\bar z_i}\Big),\frac{\partial}{\partial\bar z_k}+\phi\Big(\frac{\partial}{\partial\bar z_k}\Big)\right]\in T_t^{0,1}.$$

Since $[\frac{\partial}{\partial\bar z_i},\frac{\partial}{\partial\bar z_k}]=0$, this is equivalent to

$$\Big(\sum_l\left[\frac{\partial}{\partial\bar z_i},\phi_{kl}\frac{\partial}{\partial z_l}\right]+\sum_j\left[\phi_{ij}\frac{\partial}{\partial z_j},\frac{\partial}{\partial\bar z_k}\right]+\sum_{j,l}\left[\phi_{ij}\frac{\partial}{\partial z_j},\phi_{kl}\frac{\partial}{\partial z_l}\right]\Big)\in T_t^{0,1}.$$

Now,

$$\left[\frac{\partial}{\partial\bar z_i},\phi_{kl}\frac{\partial}{\partial z_l}\right]=\frac{\partial\phi_{kl}}{\partial\bar z_i}\frac{\partial}{\partial z_l}\ \text{ and }\ \left[\phi_{ij}\frac{\partial}{\partial z_j},\frac{\partial}{\partial\bar z_k}\right]=-\frac{\partial\phi_{ij}}{\partial\bar z_k}\frac{\partial}{\partial z_j}$$

and, hence,

$$\begin{aligned}&\sum_l\left[\frac{\partial}{\partial\bar z_i},\phi_{kl}\frac{\partial}{\partial z_l}\right]+\sum_j\left[\phi_{ij}\frac{\partial}{\partial z_j},\frac{\partial}{\partial\bar z_k}\right]\\ =&\sum_j\left(\frac{\partial\phi_{kj}}{\partial\bar z_i}-\frac{\partial\phi_{ij}}{\partial\bar z_k}\right)\frac{\partial}{\partial z_j}\\ =&(\bar\partial\phi)\Big(\frac{\partial}{\partial\bar z_i},\frac{\partial}{\partial\bar z_k}\Big).\end{aligned}$$

On the other hand,

$$\sum_{j,l}\left[\phi_{ij}\frac{\partial}{\partial z_j},\phi\frac{\partial}{\partial z_l}\right]=[\phi,\phi]=\left(\frac{\partial}{\partial\bar{z}_i},\frac{\partial}{\partial\bar{z}_k}\right).$$

Therefore, the integrability condition implies that $\bar{\partial}\phi+[\phi,\phi]\in\mathcal{A}^{0,2}(\mathcal{T}^{1,0}X)$ satisfies

$$(\bar{\partial}\phi+[\phi,\phi])\left(\frac{\partial}{\partial\bar{z}_i},\frac{\partial}{\partial\bar{z}_k}\right)\in T_t^{0,1}.$$

Hence, $\bar{\partial}+[\phi,\phi]\in\mathcal{A}^{0,2}(T^{1,0}X\cap T_t^{0,1})$. But for small t one has $T^{1,0}X\cap T_t^{0,1}=0$ and, therefore, $\bar{\partial}\phi+[\phi,\phi]=0$.

Coversely, if the Maurer-Cartan equation holds true, then integrability condition is satisfied for a local frame of $T_t^{0,1}$. This immediately shows that it is true for all sections of $T_t^{0,1}$. □

Let us now consider the power series expansion of the Maurer-Cartan equation, i.e. we replace ϕ in (1) by $\phi=\sum_{i=1}^{\infty}\phi_i t^i$:

$$\bar{\partial}\left(\sum_{i=1}^{\infty}\phi_i t^i\right)+\sum_{i,j=1}^{\infty}[\phi_i,\phi_j]t^{i+j}=0.$$

This yields a recursive system of equations:

$$\begin{aligned}0&=\bar{\partial}\phi_1\\0&=\bar{\partial}\phi_2+[\phi_1,\phi_1]\\&\vdots\\0&=\bar{\partial}\phi_k+\sum_{0<i<k}[\phi_i,\phi_{k-i}].\end{aligned}$$

We shall first study the first-order equation $\bar{\partial}\phi_1=0$ in detail. The Maurer-Cartan equation is, in particular, saying that the first-order deformation of the complex structure I is described by a $\bar{\partial}$-closed $(0,1)$-form ϕ_1 with values in the holomorphic tangent bundle $\mathcal{T}_X$. Thus, it defines an element $[\phi_1]\in H^1(X,\mathcal{T}_X)$.

Definition 1.1.3. The **Kodaira-Spencer class** of a one-parameter deformation I_t of the complex structure I is the induced cohomology class $[\phi_1]\in H^1(X,\mathcal{T}_X)$.

In order to identify isomorphic (first order) deformations, the (infinitesimal) action of the diffeomorphism group $\mathrm{Diff}(M)$ has to be taken into account. Let F_t be a one-parameter family of diffeomorphisms of the manifold M. With respect to local coordinate functions x_i this can be expanded into a power series $\tilde{x_i}=x_i+tF_i(x)+t^2\ldots$. The first order term $\sum_i F_i\frac{\partial}{\partial x_i}$ defines a global vector field

$$\frac{dF_t}{dt}|_{t=0}\in\mathcal{A}^0(TM).$$

We can rewrite this identification on the complex manifold $X=(M,I)$ withh respect to holomorphic coordinates z_i as $\tilde{z_i}=z_i+tf_i(z,\bar{z})+t^2\ldots$. For the antiholomorphic coordinates $\bar{z_i}$ we then have $\bar{\tilde{z}_i}=\bar{z_i}+t\bar{f}_i(z,\bar{z})+t^2\ldots$. Thus, $\frac{dF_t}{dt}|_{t=0}\in\mathcal{A}^0(TX)$ in local coordinates (up to the factor 2) is given as

$$\sum f_i\frac{\partial}{\partial z_i}+\sum\bar{f}_i\frac{\partial}{\partial\bar{z}_i}.$$

Any diffeomorphism can be used to construct new complex structures out of the given one I by pushing-forward I via the differential of the diffeomorphism. More explicitly, for the diffeomorphism F_t the new complex structure is given by $dF_t \circ I \circ (dF_t)^{-1}$. The antiholomorphic tangent bundle of this complex structure is thus

$$T_t^{0,1} = dF_t(T^{0,1}X).$$

In local coordinates the first order deformation of $T^{0,1}X$ is described as the image of $T^{0,1}X$ under

$$\mathrm{id} + \sum_{i,j} \frac{\partial f_i}{\partial z_j} \otimes \frac{\partial}{\partial z_j} + \frac{\partial f_i}{\partial \bar{z_j}} d\bar{z_j} \otimes \frac{\partial}{\partial z_i} + \frac{\partial \bar{f}_i}{\partial z_j} dz_j \otimes \frac{\partial}{\partial \bar{z_i}} + \frac{\partial \bar{f}_i}{\partial \bar{z_j}} d\bar{z_j} \otimes \frac{\partial}{\partial \bar{z_i}}.$$

In this sum, only the third and fifth terms are non-trivial on $T^{0,1}X$. Moreover, the fifth term is an endomorphism of $T^{0,1}X$.

Thus, the first order deformation of $T^{0,1}X$ induced by a one-parameter family of diffeomorphisms F_t of M is described by the homomorphism

$$T^{0,1}X \longrightarrow T^{1,0}X, \ \frac{\partial}{\partial \bar{z_j}} \longmapsto \sum_i \frac{\partial f_i}{\partial \bar{z_j}} \frac{\partial}{\partial z_i}.$$

This yields

Lemma 1.1.4. *The first-order deformation of the complex structure I on the manifold M (defining the complex manifold $X = (M, I)$) induced by a one-parameter family F_t of diffeomorphisms of M is determined by the map*

$$\phi_1 = \bar{\partial}\left(\left(\frac{dF_t}{dt}|_{t=0} \right)^{1,0} \right) : T^{0,1}X \longrightarrow T^{1,0}X.$$

As a consequence of this, we derive a description of all first-order deformations of isomorphism classes of complex structures of $X = (M, I)$.

Proposition 1.1.5. *Let X be a complex manifold. There is a natural bijection between all first-order deformations of X and elements of $H^1(X, \mathcal{T}_X)$.*

Proof. Write $X = (M, I)$. Then the first order deformations of I correspond to $\bar{\partial}$-closed elements in $\mathcal{A}^{0,1}(\mathcal{X})$ and by the previous lemma those that define isomorphic ones differ by elements of the image of $\bar{\partial} : \mathcal{A}^0(\mathcal{T}_X) \to \mathcal{A}^{0,1}(\mathcal{T}_X)$, i.e. $\bar{\partial}$-exact ones. As the Dolbeault cohomology just computes the sheaf cohomology of $\mathcal{T}_X$, this proves the assertion. □

In other words, the Kodaira-Spencer class determines the first-order deformations of X and any class in $H^1(X, \mathcal{T}_X)$ occurs as a Kodaira-Spencer class.

The principal task in deformation theory is to integrate given first-order deformations $v \in H^1(X, \mathcal{T}_X)$, i.e. to find a one-parameter family I_t such that its Kodaira-Spencer class is v. In general this is not possible. Obstructions may occur at any order.

Moreover, since we have to divide by the action of the diffeomorphism group at any step, we have to study all possible ϕ_i carefully. This is needed in order to ensure the convergence of the power series and to solve the recursive equations.

That there in general exist obstructions can already be guessed at order two: Once a lift ϕ_1 of $v \in H^1(X, \mathcal{T}_{\mathcal{X}})$ is chosen, we have to find ϕ_2 such that $\bar{\partial}\phi_2 = -[\phi_1, \phi_1]$. But it may very well happen that the form $[\phi_1, \phi_1]$ is never $\bar{\partial}$-exact, no matter how ϕ_1 is chosen.

Using 1.1.1 we can associate to any first order deformation $v \in H^1(X, \mathcal{T}_X)$ a class $[v, v] \in H^2(X, \mathcal{T}_X)$.

Corollary 1.1.6. *A first-order deformation $v \in H^1(X, \mathcal{T}_X)$ cannot be integrated if $[v, v] \in H^2(X, \mathcal{T}_X)$ does not vanish.*

1.2 Deformations of Calabi-Yau manifolds

Next, we will describe how the recursive system of equations given by the Maurer-Cartan equation can be sovled when the underlying manifold X admits a holomorphic volume form, i.e. when X is a **Calabi-Yau manifold**. More precisely, by a Calabi-Yau manifold we mean here a compact Kähler manifold X of dimension n with trivial canonical bundle $\bigwedge^n \Omega_X = K_X \cong \mathcal{O}_X$. Moreover, we fix a holomorphic volume form, i.e. a trivializing section $\Omega \in H^0(X, K_X)$.

The form Ω can be used to define a natural isomorphism

$$\eta : \bigwedge^p \mathcal{T}_X \cong \bigwedge^{n-p} \Omega_X.$$

One defines

$$\eta(v_1 \wedge \cdots \wedge v_p) = i_{v_1} \ldots i_{v_p}(\Omega),$$

where i_v is the contraction defined by $i_v(\alpha)(w_1, \ldots, w_i) = \alpha(v, w_1, \ldots, w_i)$. If Ω is locally written in the form $f dz_1 \wedge \cdots \wedge dz_n$, then

$$\eta(\frac{\partial}{\partial z_{i_1}} \wedge \cdots \wedge \frac{\partial}{\partial z_{i_p}}) = (-1)^{(\sum i_j)-p} f dz_1 \wedge \cdots \wedge \widehat{dz_{i_1}} \wedge \cdots \wedge \widehat{dz_{i_p}} \wedge \ldots dz_n,$$

for $i_1 < \cdots < i_p$.

Moreover, the isomorphism η induces canonical isomorphisms

$$\eta : \mathcal{A}^{0,q}(\bigwedge^p \mathcal{T}_X) \cong \mathcal{A}^{n-p,q}(X).$$

Definition 1.2.1. The operator $\Delta : \mathcal{A}^{0,q}(\bigwedge^p \mathcal{T}_X) \to \mathcal{A}^{0,q}(\bigwedge^{p-1} \mathcal{T}_X)$ is defined as

$$\Delta : \mathcal{A}^{0,q}(\bigwedge^p \mathcal{T}_X) \xrightarrow{\eta} \mathcal{A}^{n-p,q}(X) \xrightarrow{\partial} \mathcal{A}^{n-p+1,q}(X) \xrightarrow{\eta^{-1}} \mathcal{A}^{0,q}(\bigwedge^{p-1} \mathcal{T}_X).$$

Note that ∂ itself is not well-defined on $\mathcal{A}^{0,q}(\bigwedge^p \mathcal{T}_X)$.

Lemma 1.2.2. *The operator Δ anti-commutes with $\bar{\partial}$, i.e. $\Delta \circ \bar{\partial} = -\bar{\partial} \circ \Delta$.*

Proof. We use the fact that $\Omega \in \mathcal{A}^{n,0}(X)$ is holomorphic, i.e. $\bar\partial$-closed. If we write in local coordinates $\Omega = f dz_1 \wedge \cdots \wedge dz_n$, then f is a holomorphic function. Then

$$\eta(g \cdot d\bar z_I \otimes \frac{\partial}{\partial z_J}) = \pm(fg) \cdot d\bar z_I \otimes dz_{\{1,\dots,n\}\setminus J}.$$

Hence, $\bar\partial \circ \eta = \eta \circ \bar\partial$ and, therefore,

$$\begin{aligned}\bar\partial(\Delta(\alpha)) &= \eta^{-1}(\bar\partial\partial\eta(\alpha)) = -\eta^{-1}(\partial\bar\partial\eta(\alpha)) \\ &= -\eta^{-1}(\partial\eta(\bar\partial\alpha)) = -\Delta(\bar\partial(\alpha)).\end{aligned}$$

□

Also note that $\Delta^2 = 0$. Thus, Δ can be used as a differential, but it is not well-behaved with respect to the exterior product $\mathcal{A}^{0,p}(\bigwedge^r \mathcal{T}_X) \times \mathcal{A}^{0,q}(\bigwedge^s \mathcal{T}_X) \to \mathcal{A}^{0,p+q}(\bigwedge^{r+s} \mathcal{T}_X)$. This is made more precise by the following key result due to Tian and Todorov [1, 2].

Lemma 1.2.3 (Tian-Todorov lemma)**.** *If $\alpha \in \mathcal{A}^{0,p}(\mathcal{T}_X)$ and $\beta \in \mathcal{A}^{0,q}(\mathcal{T}_X)$, then*

$$(-1)^p[\alpha,\beta] = \Delta(\alpha\wedge\beta) - \Delta(\alpha)\wedge\beta - (-1)^{p+1}\alpha\wedge\Delta(\beta).$$

Proof. As the claim is local and additive in α and β, it suffices to prove it for $\alpha = a d\bar z_I \otimes \frac{\partial}{\partial z_i}$ and $\beta = b d\bar z_J \otimes \frac{\partial}{\partial z_j}$.

Then, by definition, $[\alpha,\beta] = d\bar z_I \wedge d\bar z_J [a\frac{\partial}{\partial z_i}, b\frac{\partial}{\partial z_j}]$.

In a first step we shall reduce the assertion to the case $p = q = 0$. Using the short hand

$$G(\alpha,\beta) := \Delta(\alpha\wedge\beta) - \Delta(\alpha)\wedge\beta - (-1)^{p+1}\alpha\wedge\Delta(\beta),$$

one does this by proving

$$G(\alpha,\beta) = (-1)^p(d\bar z_I \wedge d\bar z_J) \otimes G(a\frac{\partial}{\partial z_i}, b\frac{\partial}{\partial z_j}).$$

Using the standard sign rules, one indeed computes

$$\begin{aligned}\Delta(\alpha\beta) = \eta^{-1}\partial\eta(\alpha\wedge\beta) &= (-1)^q\eta^{-1}\partial\eta\Big((d\bar z_I \wedge d\bar z_J) \otimes (a\frac{\partial}{\partial z_i} \wedge b\frac{\partial}{\partial z_j})\Big) \\ &= (-1)^q\eta^{-1}\partial\Big((d\bar z_I \wedge d\bar z_J) \otimes \eta\Big(a\frac{\partial}{\partial} \wedge b\frac{\partial}{\partial z_j}\Big)\Big) \\ &= (-1)^q \cdot (-1)^{p+q}\eta^{-1}\Big((d\bar z_I \wedge d\bar z_J) \otimes \partial\eta\Big(a\frac{\partial}{\partial z_i} \wedge b\frac{\partial}{\partial z_j}\Big)\Big). \\ &= (-1)^p(d\bar z_I \wedge d\bar z_J) \otimes \Delta\Big(a\frac{\partial}{\partial z_i} \wedge b\frac{\partial}{\partial z_j}\Big).\end{aligned}$$

Similarly,

$$\begin{aligned}\Delta(\alpha)\wedge\beta = \eta^{-1}\partial\eta\Big(d\bar z_I \otimes a\frac{\partial}{\partial z_i}\Big)\wedge\beta &= (-1)^p\Big(d\bar z_I \otimes \Delta\Big(a\frac{\partial}{\partial z_i}\Big)\Big)\wedge\beta \\ &= (-1)^p(d\bar z_I \wedge d\bar z_J) \otimes \Big(\Delta\Big(a\frac{\partial}{\partial z_i}\Big) \wedge b\frac{\partial}{\partial z_j}\Big)\end{aligned}$$

and

$$(-1)^{p+1}\alpha\wedge\Delta(\beta)=(-1)^{p+1}\alpha\wedge\Big((-1)^q d\bar z_J\otimes\Delta\Big(b\frac{\partial}{\partial z_j}\Big)\Big)$$
$$=(-1)^p(-1)(d\bar z_I\wedge d\bar z_J)\otimes\Big(a\frac{\partial}{\partial z_i}\wedge\Delta(b\frac{\partial}{\partial z_j})\Big).$$

Thus,

$$\begin{aligned}G(\alpha,\beta)=&(-1)^p(d\bar z_I\wedge d\bar z_J)\otimes\Big(\Delta\Big(a\frac{\partial}{\partial z_i}\wedge b\frac{\partial}{\partial z_j}\Big)\\&-\Delta\Big(a\frac{\partial}{\partial z_i}\Big)\wedge b\frac{\partial}{\partial z_j}-(-1)a\frac{\partial}{\partial z_i}\wedge\Delta\Big(b\frac{\partial}{\partial z_j}\Big)\Big)\\=&(-1)^p(d\bar z_I\wedge d\bar z_J)\otimes G\Big(a\frac{\partial}{\partial z_i},b\frac{\partial}{\partial z_j}\Big).\end{aligned}$$

In order to treat the case $p=q=0$, let us first mention explicitly

$$\eta(\frac{\partial}{\partial z_i})=(-1)^{i-1}f dz_1\wedge\cdots\wedge\widehat{dz_i}\wedge\cdots\wedge dz_n$$

and

$$\eta\Big(\frac{\partial}{\partial z_i}\wedge\frac{\partial}{\partial z_j}\Big)=(-1)^{i+j-2}f dz_1\wedge\cdots\wedge\widehat{dz_i}\wedge\cdots\wedge\widehat{dz_j}\wedge\cdots\wedge dz_n.$$

This also shows $\eta^{-1}(dz_1\wedge\cdots\wedge\widehat{dz_i}\wedge\dots dz_n)=f^{-1}\frac{\partial}{\partial z_i}$ and $\eta(dz_1\wedge\cdots\wedge dz_n)=f^{-1}$.

Hence,

$$\begin{aligned}\Delta(a\frac{\partial}{\partial z_i})&=\eta^{-1}\partial\Big((-1)^{i-1}(af)dz_1\wedge\cdots\wedge\widehat{dz_i}\wedge\cdots\wedge dz_n\Big)\\&=\eta^{-1}\frac{\partial(af)}{\partial z_i}dz_1\wedge\cdots\wedge dz_n=\frac{\partial(af)}{\partial z_i}f^{-1}\end{aligned}$$

and similarly

$$\Delta\Big(b\frac{\partial}{\partial z_j}\Big)=\frac{\partial(bf)}{\partial z_j}f^{-1}.$$

Furthermore,

$$\begin{aligned}\Delta\Big(a\frac{\partial}{\partial z_i},b\frac{\partial}{\partial z_j}\Big)=&\eta^{-1}\partial\Big((-1)^{i+j}(abf)dz_1\wedge\cdots\wedge\widehat{dz_i}\wedge\cdots\wedge\widehat{dz_j}\wedge\cdots\wedge dz_n\Big)\\=&\eta^{-1}\Big((-1)^{j-1}\frac{\partial(abf)}{\partial z_i}dz_1\wedge\cdots\wedge\widehat{dz_j}\wedge\cdots\wedge dz_n\\&+(-1)^i\frac{\partial(abf)}{\partial z_j}dz_1\wedge\cdots\wedge\widehat{dz_i}\wedge\cdots\wedge dz_n\Big)\\=&\frac{\partial(abf)}{\partial z_i}f^{-1}\frac{\partial}{\partial z_j}-\frac{\partial(abf)}{\partial z_j}f^{-1}\frac{\partial}{\partial z_i}\\=&\Big[a\frac{\partial}{\partial z_i},b\frac{\partial}{\partial z_j}\Big]+\Delta\Big(a\frac{\partial}{\partial z_i}\Big)b\frac{\partial}{\partial z_j}-a\frac{\partial}{\partial z_i}\Delta\Big(b\frac{\partial}{\partial z_j}\Big).\end{aligned}$$

This proves the assertion for $p=q=0$. □

Corollary 1.2.4. *Let α and β be Δ-closed elements of $\mathcal{A}^{0,*}(\mathcal{T}_X)$. Then the bracket $[\alpha,\beta]$ is Δ-exact.*

The Tian-Todorov lemma will be applied to the coefficients $\phi_i \in \mathcal{A}^{0,1}(\mathcal{T}_X)$ of the power series expansion of a deformation of the complex structure. Before proving the general result, we shall spell out the first two steps.

Let $v \in H^1(X, \mathcal{T}_X)$ be given. We first look for a good representative $\phi_1 \in \mathcal{A}^{0,1}(\mathcal{T}_X)$. In order to solve the Maurer-Cartan equation $\bar{\partial}\phi_2 = -[\phi_1, \phi_1]$, we have to ensure that

- $[\phi_1, \phi_1]$ is $\bar{\partial}$-closed
- $[\phi_1, \phi_1]$ has no harmonic part.

Here we use the Hodge decomposition $\mathcal{A}^{0,1}(\mathcal{T}_X) = \bar{\partial}\mathcal{A}^0(\mathcal{T}_X) \oplus \mathcal{H}^{0,1}(\mathcal{T}_X) \oplus \bar{\partial}^*\mathcal{A}^{0,2}(\mathcal{T}_X)$ with respect to an hermitian metric on X and thus on $\mathcal{T}_X$.

The first condition is always satisfied: If $\alpha_i \in \mathcal{A}^{0,q_i}(\mathcal{T}_X)$, $i = 1, 2$, are $\bar{\partial}$-closed, then so is $[\alpha_1, \alpha_2]$. This follows from (1.1.1).

The Tian-Todorov lemma yields: If $\alpha_i \in \mathcal{A}^{0,q_i}(\mathcal{T}_X)$, $i = 1, 2$ are chosen such that $\eta(\alpha_i) \in \mathcal{A}^{n-1,q_i}(X)$ are ∂-closed, then $\eta[\alpha_1, \alpha_2]$ is ∂-exact. More explicitly, $\eta[\alpha_1, \alpha_2] = (-1)^{q_1} \cdot \partial\eta(\alpha_1 \wedge \alpha_2)$.

Eventually, we find: If $\phi_1 \in \mathcal{A}^{0,1}(\mathcal{T}_X)$ is chosen such that $\eta(\phi_1) \in \mathcal{A}^{n-1,1}(X)$ is harmonic with respect to a Kähler metric on X, then $\bar{\partial}\phi_2 = -[\phi_1, \phi_1]$ admits a solution $\phi_2 \in \mathcal{A}^{0,1}(\mathcal{T}_X)$ which can, moreover, be chosen such that $\eta(\phi_2)$ is ∂-exact.

Indeed, by the above arguments $\eta[\phi_1, \phi_1]$ is ∂-exact and $\bar{\partial}$-closed. Since by Hodge theory on a compact Kähler manifold a ∂-exact form cannot have a non-trivial harmonic part, this proves that $\eta[\phi_1, \phi_1]$ is $\bar{\partial}$-exact. But using the $\partial\bar{\partial}$-lemma, we can, moreover, say that there exists a form $\gamma \in \mathcal{A}^{n-2,1}(X)$ with $\bar{\partial}\partial\gamma = -\eta[\phi_1, \phi_1]$. Then set $\phi_2 := \eta^{-1}(\partial\gamma)$. Since η commutes with $\bar{\partial}$, this proves the claim.

All these arguments can now easily be generalized to prove the general result, which is also due to Tian and Todorov.

Proposition 1.2.5. *Let X be a Calabi-Yau manifold and let $v \in H^1(X, \mathcal{T}_X)$. Then there exists a formal power series $\phi_1 t + \phi_2 t^2 + \cdots$ with $\phi_i \in \mathcal{A}^{0,1}(\mathcal{T}_X)$ satisfying the Maurer-Cartan equations*

$$\bar{\partial}\phi_1 = 0 \text{ and } \bar{\partial}\phi_k = -\sum_{0<i<k} [\phi_i, \phi_{k-i}],$$

with $[\phi_1] = v$ and such that $\eta(\phi_i) \in \mathcal{A}^{n-1,1}(X)$ is ∂-exact for all $i > 1$.

Proof. We begin with $\phi_1 \in \mathcal{A}^{0,1}(\mathcal{T}_X)$ representing v such that $\eta(\phi_1) \in \mathcal{A}^{n-1,1}(X)$ is harmonic. Then one finds ϕ_2 as above. Let us suppose that we have found $\phi_2, \ldots, \phi_{k-1}$ as claimed by the proposition. Then $\eta[\phi_i, \phi_{k-i}]$ is ∂-exact for $0 < i < k$ by the Tian-Todorov lemma. Thus, in order to find ϕ_k satisfying the assertion, we have to ensure that the sum $\sum_{0<i<k}[\phi_i, \phi_{k-i}]$ is $\bar{\partial}$-closed. As we have used implicitly before, one has

$$\bar{\partial}\Big(\sum_{0<i<k} [\phi_i, \phi_{k-i}]\Big) = \sum_{0<i<k} ([\bar{\partial}\phi_i, \phi_{k-i}] - [\phi_i, \bar{\partial}\phi_{k-i}])$$

and by induction hypothesis this equals

$$\begin{aligned}
&-\sum_{0<i<k}\left(\sum_{0<j<i}[[\phi_j,\phi_{i-j}],\phi_{k-i}]-\sum_{0<l<k-i}[\phi_i,[\phi_l,\phi_{k-i-l}]]\right)\\
=&-\sum_{0<i<k}\sum_{0<j<i}[[\phi_j,\phi_{i-j}],\phi_{k-i}]+\sum_{0<i<k}\sum_{0<l<i}[\phi_{k-i},[\phi_l,\phi_{i-l}]]\\
=&2\sum_{0<i<k}\sum_{0<l<i}[\phi_{k-i},[\phi_l,\phi_{i-l}]]=0
\end{aligned}$$

where we use $[\alpha,\beta]=-[\beta,\alpha]$ for $\alpha\in\mathcal{A}^{0,2}(\mathcal{T}_X)$ and arbitrary β and by Jacobi identity we deduce that the final result is 0.

The $\partial\bar{\partial}$-lemma eventually ensures that we can choose ϕ_k of the form $\eta^{-1}(\partial\gamma)$, i.e. $\eta(\phi_k)$ is ∂-exact. □

Remark 1.2.6. There might be very well other solutions of the Maurer-Cartan equation which do not satisfy the extra condition that $\eta(\phi_i)$ is ∂-exact for $i>1$. In fact, the $\eta(\phi_i)$ need not even be ∂-closed.

Remark 1.2.7. So far, we have not said anything about the existence of a convergent solution. In fact, there is a standard procedure to turn any formal solution into a convergent one. At this point, a good deal of analysis arises. For the details we refer to [3, Sect. 5.3].

The principal idea is to consider the operator $\bar{\partial}$ on $\mathcal{A}^{0,q}(\mathcal{T}_X)$ and its adjoint operator with respect to a chosen Kähler (or just hermitian) metric. For clarity, let us write $\bar{\partial}_{\mathcal{T}_X}$ respectively $\bar{\partial}^*_{\mathcal{T}_X}$ for these two operators. Then one shows that a formal solution converges if the coefficients ϕ_k are $\bar{\partial}^*_{\mathcal{T}_X}$-exact. It is then not too difficult to transform any formal solution into one that satisfies this additional condition, as the mere existence of a formal solution ensures that all possible obstructions are trivial.

2 Deformation theory with algebraic approaches

2.1 The T^i functors

We first present the construction and main properties of the T^i functors introduced by Lichtenbaum and Schlessinger [4].

Let $A\to B$ be a homomorphism of rings and let M be a B-module. Here we construct groups $T^i(B/A,M)$ for $i=0,1,2$. The rings are assumed to be commutative with identity, but we do not impose any finiteness conditions yet.

First choose a polynomial ring $R=A[x]$ in a set of variables $x=\{x_i\}$ (possibly infinite) such that B can be written as a quotient of R as an A-algebra. Let I be the ideal defining B, so that there is an exact sequence

$$0\longrightarrow I\longrightarrow R\longrightarrow B\longrightarrow 0.$$

Second choose a free R-module F and a surjection $j:F\to I\to 0$ and let Q be the kernel:

$$0\longrightarrow Q\longrightarrow F\xrightarrow{j} I\longrightarrow 0.$$

Let

$$F_0 = (j(a)b - j(b)a)_{a,b\in F} \subseteq F.$$

Define a complex of B-modules, called the **cotangent complex**

$$L_2 \xrightarrow{d_2} L_1 \xrightarrow{d_1} L_0$$

as follows.

- Take $L_2 = Q/F_0$, which is a B-module. Indeed, if $x \in I$ and $a \in Q$. Write $x = j(x')$ for some $x' \in F$, then
$$xa = j(x')a \equiv j(a)x' = 0 \pmod{F_0}.$$
- Take $L_1 = F \otimes_R B = F/IF$, and let $d_2 : L_2 \to L_1$ be the map induced from the inclusion $Q \to F$.
- Take $L_0 = \Omega_{R/A} \times_R B$, where $\Omega_{R/A}$ is the module of relative Kähler differential. Let d_1 be the composition of
$$\begin{aligned} L_1 \twoheadrightarrow I/I^2 &\to \Omega_{R/A} \otimes_R B \\ x &\to d_{R/A}(x) \otimes 1 \end{aligned}$$
It is clear that $d_1 d_2 = 0$, so we defined a complex of B-modules. Notice that L_0, L_1 are free B-modules, since R is a polynomial ring over A.
- Indeed, the second fundamental exact sequence of Kähler differential
$$I/I^2 \xrightarrow{d} \Omega_{R/A} \otimes_R B \longrightarrow \Omega_{B/A} \longrightarrow 0$$
give the exact sequence
$$L_2 \xrightarrow{d_2} L_1 \xrightarrow{d_1} L_0 \longrightarrow \Omega_{B/A} \longrightarrow 0. \tag{2}$$

Lemma 2.1.1. *For any B-module M, define*

$$T^i(B/A, M) := H^i(\mathrm{Hom}_B(L_\bullet, M)),$$

which is independent of the choice of F and R.

Proof. First, we fixed the polynomial R. If F and F' are two choices of free R-modules mapping onto I, then $F \oplus F'$ gives the third choice. Since F' is free, the map $j' : F' \to I$ factors through F, i.e., $j' = jp$ for some map $p : F' \to F$. Replacing each generator e' of $F' \subseteq F \oplus F'$ by $e' - p(e')$, we may assume that the map $F \oplus F' \to I$ is just $(j, 0)$. Then $\ker(j, 0) = Q \oplus F'$ and thus $(F \oplus F')_0 = F_0 + IF'$. Let $L'_\bullet$ be the complex obtained from $(F \oplus F', R)$, then

$$L'_2 = L_2 \oplus F'/IF',\ L'_1 = L_1 \oplus (F' \otimes_R B),\ L'_0 = L_0.$$

Hence, $L'_\bullet$ is obtained by the direct sum of $L_\bullet$ with the free acyclic complex $F' \otimes_R B \to F' \otimes_R B \to 0$, which induces same cohomology after taking Hom functor.

Second, it suffices to compare $R = A[x]$ with $R' = A[x, y]$. Again, the map $A[y] \to B$ can be factored through $A[x]$ by a homomorphism $p : A[y] \to A[x]$. Replacing each $y_i \in A[x, y]$ by $y_i - p(y_i)$, we may assume the ring homomorphism $A[x, y] \to B$ maps y_i to 0, then it has kernel $IR' + yR'$. Since T^i is independent of the choice of F, we may take $F' = F \otimes_R R'$ and G be a free R'-module on the index set of the y variables. Then we have the following diagram

$$\begin{array}{ccccccccc} 0 & \longrightarrow & Q' & \longrightarrow & F' \oplus G' & \longrightarrow & IR' \oplus yR' & \longrightarrow & 0 \\ & & \uparrow & & \uparrow & & \uparrow & & \\ 0 & \longrightarrow & Q & \longrightarrow & F & \longrightarrow & I & \longrightarrow & 0 \end{array}$$

and it is clear that

$$Q' = QR' + \langle y_i a - j(a)e_i \mid a \in F \rangle_{R'} + \langle y_i e_j - y_j e_i \rangle_{R'}.$$

Since $\langle y_i a - j(a)e_i \mid a \in F \rangle_{R'} + \langle y_i e_j - y_j e_i \rangle_{R'} \subseteq (F' \oplus G')_0$, we have $L'_2 = L_2$.

On the other hand, $L'_1 = L_1 \oplus (G' \otimes_{R'} B)$ and $L'_0 = L_0 \oplus (\Omega_{A[y]/A} \otimes B)$. Then $L'_\bullet$ is obtained by the direct sum of $L_\bullet$ with the free acyclic complex

$$\begin{aligned} G' \otimes_{R'} B &\longrightarrow \Omega_{A[y]/A} \otimes B \longrightarrow 0 \\ e_i &\longrightarrow dy_i, \end{aligned}$$

and thus induces same cohomology after taking Hom functor. □

Theorem 2.1.2. *Let $A \to B$ be a homomorphism of rings. Then $T^i(B/A, -)$ is a covariant, additive functor from Mod_B to itself. If*

$$0 \longrightarrow M' \longrightarrow M \longrightarrow M'' \longrightarrow 0$$

is a short exact sequence of B-modules, then there is a long exact sequence

$$\begin{aligned} 0 \longrightarrow T^0(B/A, M') &\longrightarrow T^0(B/A, M) \longrightarrow T^0(B/A, M'') \\ \longrightarrow T^1(B/A, M') &\longrightarrow T^1(B/A, M) \longrightarrow T^1(B/A, M'') \\ \longrightarrow T^2(B/A, M') &\longrightarrow T^2(B/A, M) \longrightarrow T^2(B/A, M''). \end{aligned}$$

Proof. By construction, $T^i(B/A, -)$ is a covariant additive functor. Since the terms L_1 and L_0 of the complex $L_\bullet$ are free, we get a sequence of complexes

$$0 \longrightarrow \mathrm{Hom}_B(L_\bullet, M') \longrightarrow \mathrm{Hom}_B(L_\bullet, M) \longrightarrow \mathrm{Hom}_B(L_\bullet, M'') \longrightarrow 0$$

that is exact except possibly for the map

$$\mathrm{Hom}_B(L_2, M) \to \mathrm{Hom}_B(L_2, M'')$$

may not be surjective. This sequence of complexes gives the long exact sequence of cohomology above. □

Theorem 2.1.3. *Let $A \to B \to C$ be homomorphisms of rings, and let M be a C-module. Then there is a long exact sequence*

$$\begin{aligned} 0 &\longrightarrow T^0(C/B, M) \longrightarrow T^0(C/A, M) \longrightarrow T^0(B/A, M) \\ &\longrightarrow T^1(C/B, M) \longrightarrow T^1(C/A, M) \longrightarrow T^1(B/A, M) \\ &\longrightarrow T^2(C/B, M) \longrightarrow T^2(C/A, M) \longrightarrow T^2(B/A, M). \end{aligned}$$

Proof. Choose

$$0 \longrightarrow I \longrightarrow A[x] \longrightarrow B \longrightarrow 0, \quad 0 \longrightarrow Q \longrightarrow F \xrightarrow{j} I \longrightarrow 0$$

$$0 \longrightarrow J \longrightarrow B[y] \longrightarrow C \longrightarrow 0, \quad 0 \longrightarrow P \longrightarrow G \longrightarrow J \longrightarrow 0$$

to calculate $T^i(B/A, M)$ and $T^i(C/B, M)$. Then $A[x, y] \to B[y] \to C$ is surjective, and denote its kernel by K. It induces an exact sequence

$$0 \longrightarrow I[y] \longrightarrow K \longrightarrow J \longrightarrow 0$$

of A-modules. Let F' and G' be free $A[x, y]$-modules on the same index sets as F and G respectively. Choose a lifting of the map $G \to J$ to a map $G' \to K$. From the map $F \to I$ we get the natural map $F' \to I[y]$. Hence, we get the surjection $F' \oplus G' \to K$, and let S be its kernel. Now, we can calculate $T^i(C/A, M)$ by

$$0 \longrightarrow K \longrightarrow A[x, y] \longrightarrow C \longrightarrow 0, \quad 0 \longrightarrow S \longrightarrow F' \oplus G' \xrightarrow{j} K \longrightarrow 0.$$

Now, we have the following diagram

$$\begin{array}{ccccccccc} 0 & \longrightarrow & Q[y] & \longrightarrow & F' & \longrightarrow & I[y] & \longrightarrow & 0 \\ & & \downarrow & & \downarrow & & \downarrow & & \\ 0 & \longrightarrow & S & \longrightarrow & F' \oplus G' & \longrightarrow & K & \longrightarrow & 0 \\ & & \downarrow & & \downarrow & & \downarrow & & \\ 0 & \longrightarrow & P & \longrightarrow & G & \longrightarrow & J & \longrightarrow & 0 \end{array}$$

which is exact for each horizontal sequence, but not exact for vertical sequence. It induces the maps of complexes

$$L_\bullet(B/A) \otimes_B C \to L_\bullet(C/A) \to L_\bullet(C/B).$$

Indeed, on the degree 0 level we have split exact sequence.

$$\Omega_{A[x]/A} \otimes_{A[x]} C \longrightarrow \Omega_{A[x,y]/A} \otimes_{A[x,y]} C \longrightarrow \Omega_{B[y]/B} \otimes_{B[y]} C$$

$$dx_i \longmapsto dx_i$$

$$dx_i,\ dy_j \longmapsto 0,\ dy_j$$

On the degree 1 level we have split exact sequence

$$F \otimes_{A[x]} C \longrightarrow (F' \oplus G') \otimes_{A[x,y]} C \longrightarrow G \otimes_{B[y]} C$$

of free C-modules. On the degree 2 level we have

$$(Q/F_0) \otimes_B C \longrightarrow S/(F' \oplus G')_0 \longrightarrow P/G_0. \tag{3}$$

We claim that $S \to P$ is surjective. Indeed, if $p \in P$. Take $(f', g') \in F' \oplus G'$ mapping to p. Then the image of (f', g') in K is contained in $I[y]$. Since $F' \to I[y]$ is surjective, there exists $\tilde{f} \in F'$ having the same image in K as (f', g'). Hence, $(f' - \tilde{f}, g') \in S$ maps to p.

To complete the proof, we remain to show the exactness of (3) in the middle. Let $s = f + g$ be an element of $S \subseteq F' \oplus G'$ such that its image in P is contained in G_0. Then the image g of g' in G belongs to G_0, write $g = j(a)b - j(b)a$ for some $a, b \in G$. Lift a, b to elements a', b' in G', then $j(a')b' - j(b')a' \in S$. Let $g'' = g - j(a')b' + j(b')a'$, then $g'' \in \ker(G' \to G) = IG'$. Say $g'' = xh$ and $x = j(x')$ for some $x \in I$, $h \in G'$, and $x' \in F$. Then

$$xh = j(x')h \equiv j(h)x' \pmod{(F' + G')_0}.$$

Therefore,

$$s \equiv f' + g'' \equiv f' + j(h)x' \pmod{(F' + G')_0}$$

and $f' + j(h)x' \in F' \cap S = Q[y]$. □

Proposition 2.1.4. *For any morphism $A \to B$ and $M \in \mathsf{Mod}_B$,*

$$T^0(B/A, M) = \mathrm{Hom}_B(\Omega_{B/A}, M) = \mathrm{Der}_A(B, M).$$

In particular,

$$T^0(B/A, B) = \mathrm{Hom}_B(\Omega_{B/A}, B) = T_{B/A}$$

is the tangent module of B over A.

Proof. Taking $\mathrm{Hom}(-, M)$ to the exact sequence (2), we have the exact sequence

$$0 \to \mathrm{Hom}(\Omega_{B/A}, M) \to \mathrm{Hom}(L_0, M) \to \mathrm{Hom}(L_1, M).$$

□

Proposition 2.1.5. *If B is a polynomial ring over A, then $T^i(B/A, M) = 0$ for $i = 1, 2$ and for all M.*

Proof. We may take $R = B$ and $F = 0$ in our construction. Hence, the cotangent complex only has L_0 term. □

Proposition 2.1.6. *If $A \to B$ is a surjective ring homomorphism with kernel I, then*

$$T^0(B/A, M) = 0,\ T^1(B/A, M) = \mathrm{Hom}_B(I/I^2, M)$$

for any $M \in \mathsf{Mod}_B$. In particular, $T^1(B/A, B) = \mathrm{Hom}_B(I/I^2, B)$ is the normal module $N_{B/A}$ of $\mathrm{Spec}\, B$ *in* $\mathrm{Spec}\, A$.

Proof. In this case, we can take $R = A$, and thus $L_0 = 0$. Hence, $T^0 = 0$ for any module M. Furthermore, the exact sequence

$$0 \longrightarrow Q \longrightarrow F \xrightarrow{j} I \longrightarrow 0$$

tensored with B gives

$$\begin{array}{ccccccc} Q \otimes_A B & \longrightarrow & F \otimes_A B & \xrightarrow{j} & I/I^2 & \longrightarrow & 0 \\ \downarrow & & \downarrow \sim & & & & \\ Q/F_0 & \longrightarrow & F/IF & & & & \end{array}$$

Hence, we have an exact sequence

$$L_2 \longrightarrow L_1 \longrightarrow I/I^2 \longrightarrow 0,$$

and thus $T^1(B/A, M) = \mathrm{Hom}_B(I/I^2, M)$. □

Corollary 2.1.7. *Suppose that A is a local ring and $B = A/I$, where I is generated by a regular sequence $a_1, \ldots, a_r$. Then $T^2(B/A, M) = 0$ for all M.*

Proof. Since the Koszul complex of a regular sequence is exact, we conclude that $Q = F_0$. Hence, $L_2 = 0$ and $T^2(B/A, M) = 0$ for all M. □

Proposition 2.1.8. *Suppose that $A = k[x_1, \ldots, x_n]$ and $B = A/I$. Then for any M, there is an exact sequence*

$$0 \longrightarrow T^0(B/k, M) \longrightarrow \mathrm{Hom}_A(\Omega_{A/k}, M) \longrightarrow \mathrm{Hom}_B(I/I^2, M) \longrightarrow T^1(B/k, M) \longrightarrow 0$$

and an isomorphism

$$T^2(B/A, M) \xrightarrow{\sim} T^2(B/k, M).$$

Proof. It follows from the long exact sequence of T^i-functors for $k \to A \to B$ and (2.1.4, 2.1.5, 2.1.6). □

Remark 2.1.9. We have not made any finiteness assumption on the rings and modules. It is clear that if A is a noetherian ring, B is a finitely generated A-algebra, and M is a finitely generated B-module, then $T^i(B/A, M)$ are finitely generated B-modules.

Lemma 2.1.10. *The construction of the T^i functors is compatible with localization, and thus we may define sheaves $\mathcal{T}^i(X/Y, \mathcal{F})$ for any morphism of schemes $f : X \to Y$ and any sheaf $\mathcal{F}$ of $\mathcal{O}_X$-modules, such that for any open affine $\mathrm{Spec}\, A \subseteq Y$ and any open affine $\mathrm{Spec}\, B \subseteq f^{-1}(\mathrm{Spec}\, A)$, where $\mathcal{F}|_{\mathrm{Spec}\, B} = \widetilde{M}$, the section $\mathcal{T}^i(X/Y, \mathcal{F})(\mathrm{Spec}\, B) = T^i(B/A, M)$.*

For convenience, denote the modules $T^i(B/A, B)$ and $T^i(B/k, B)$ by $T^i_{B/A}$ and $T^i_{B/k}$ (or T^i_B) if no confusion arises. Furthermore, $T^0_{B/A}$ will be written as the tangent module $T_{B/A}$ of B over A. Similarly for the sheaves $\mathcal{T}^i(X/Y, \mathcal{O}_X)$ and $\mathcal{T}^i(X/k, \mathcal{O}_X)$ we write $\mathcal{T}^i_{X/Y}$ and $\mathcal{T}^i_{X/k}$ (or $\mathcal{T}^i_X$). The sheaf $\mathcal{T}^0_X$ will be written as the tangent sheaf $\mathcal{T}_X$ of X.

Proposition 2.1.11 (Base change)**.** *Assume that A is Noetherian and B is a finitely generated A-algebra.*

(1) Let $M \in \mathsf{Mod}_B$ and $A \to A'$ be a flat morphism. Let $B' = B \otimes_A A'$ and $M' = M \otimes_B B'$. Then $T^i(B/A, M) \otimes_A A' \simeq T^i(B'/A', M')$ for each i.

(2) Let $A \to A'$ be an arbitrary ring homomorphism and B is flat over A. Let $B' = B \otimes_A A'$ and $M' \in \mathsf{Mod}_{B'}$. Then $T^i(B/A, M') = T^i(B'/A', M')$ for each i.

Proof. [5, Ex. 3.7, 3.8] □

An **infinitesimal thickening** $Y \subseteq Y'$ means that Y is a closed subscheme of another scheme Y', and that the ideal $\mathcal{I}$ defining Y inside Y' is nilpotent.

Proposition 2.1.12 (Infinitesimal Lifting Property)**.** *Let X be a nonsingular affine scheme of finite type over k, let $f : Y \to X$ be a morphism from an affine scheme Y over k, and let $Y \subseteq Y'$ be an infinitesimal thickening of Y. Then the morphism f lifts to a morphism $g : Y' \to X$ such that $g|_Y = f$.*

Proof. First we note that Y' is also affine [6, III, Ex. 3.1], so we may transfer the problem to algebraic setting. Let $f : A \to B$ be the ring homomorphism over k corresponding to $f : Y \to X$ where $B = B'/I$ with $I^n = 0$ for some n. We want to find a homomorphism $g : A \to B'$ lifting f.

Consider the sequence $B' = B'/I^n \to B'/I^{n-1} \to \cdots \to B'/I^2 \to B'/I$, it will suffice to lift one step at a time. So we may assume that $I^2 = 0$.

Since X is of finite type over k, we may write $A = k[x_1, \ldots, x_n]/J$. Consider any lifting $h : k[x_1, \ldots, x_n] \to B'$ of $k[x_1, \ldots, x_n] \to A \to B$. Then h induces a map $J \to I$, and thus induces $\overline{h} : J/J^2 \to I$ since $I^2 = 0$.

Embed $X \hookrightarrow \mathbb{A}^n_k$ by $P = k[x_1, \ldots, x_n] \to A$. Since X is smooth, we obtain an exact sequence

$$0 \longrightarrow J/J^2 \longrightarrow \Omega^1_{P/k} \otimes_P A \longrightarrow \Omega^1_{A/k} \longrightarrow 0.$$

Notice that those modules correspond to locally free sheaves on X, and thus are projective A-modules. Via the maps h, f, we get a P-module structure on B', and A-module structures on B, I. Applying $\mathrm{Hom}_A(-, I)$ will get another exact sequence

$$0 \longrightarrow \mathrm{Hom}_A(\Omega^1_{A/k}, I) \longrightarrow \mathrm{Hom}_P(\Omega^1_{P/k}, I) \longrightarrow \mathrm{Hom}_A(J/J^2, I) \longrightarrow 0.$$

Let $\theta \in \mathrm{Hom}_P(\Omega^1_{P/k}, I) = \mathrm{Der}_k(P, I)$ be an element whose image is $\overline{h} \in \mathrm{Hom}_A(J/J^2, I)$. We claim that $h' := h - \theta : P \to B'$ is a ring homomorphism lifting f with $h'(J) = 0$. The first assertion follows by the following lemma.

Lemma 2.1.13. *Let $B' \to B$ be a surjective homomorphism of k-algebras with kernel I of square zero. Let $R \to B$ be a homomorphism of k-algebras.*

(1) If $f, g : R \to B'$ are two liftings of the map $R \to B$ to B', then $\theta = g - f$ is a k-derivation of R to I.

(2) Conversely, if $f : R \to B'$ is one lifting, and $\theta : R \to I$ is a derivation, then $g = f + \theta$ is another homomorphism of R to B' lifting the given map $R \to B$.

In other words, if it is nonempty, the set of liftings $R \to B$ to k-algebra homomorphisms of $R \to B'$ is a principal homogeneous space under the action of $\mathrm{Der}_k(R, I) = \mathrm{Hom}_R(\Omega_{R/k}, I)$.

Proof. (1) Let $f, g : R \to B'$ and let $\theta = g - f$. As a k-linear map, θ followed by the projection $B' \to B$ is zero, so θ sends R to I. Let $x, y \in R$. Then

$$\begin{aligned}\theta(xy) &= g(xy) - f(xy)\\ &= g(x)g(y) - f(x)f(y)\\ &= g(x)(g(y) - f(y)) + f(y)(g(x) - f(x))\\ &= g(x)\theta(y) + f(y)\theta(x)\\ &= x\theta(y) + y\theta(x),\end{aligned}$$

the last step being because $g(x)$ and $f(y)$ act in I just like x, y. Thus θ is a k-derivation of R to I.

(2) Conversely, given f and θ as above, let $g = f + \theta$. Then

$$\begin{aligned}g(xy) &= f(xy) + \theta(xy)\\ &= f(x)f(y) + x\theta(y) + y\theta(x)\\ &= (f(x) + \theta(x))(f(y) + \theta(y))\\ &= g(x)g(y),\end{aligned}$$

where we note that $\theta(x)\theta(y) = 0$, since $I^2 = 0$. Thus g is a homomorphism of $R \to B'$ lifting R to B. □

Since $h'(J) \subseteq I$, we have $h'(J^2) \subseteq I^2 = 0$. Then h' descends to $P/J^2 \to B'$, and thus $h'|_{J/J^2} = h - h = 0$ by the choice of θ. Hence, h' descends to the desired homomorphism $g : A \to B'$ lifting f. □

Next, we investigate the relation between nonsingularity and the T^i functors.

Theorem 2.1.14. *Let $X = \operatorname{Spec} B$ be an affine scheme over an algebraically closed field k. Then X is nonsingular if and only if $T^1(B/k, M) = 0$ for all $M \in \mathsf{Mod}_B$. Furthermore, if X is nonsingular, then $T^2(B/k, M) = 0$ for all M.*

Proof. Write B as a quotient of a polynomial ring $A = k[x_1, \ldots, x_n]$ over k. Then X is nonsingular if and only if the conormal sequence

$$0 \longrightarrow I/I^2 \longrightarrow \Omega_{A/k} \otimes_A B \longrightarrow \Omega_{B/k} \longrightarrow 0 \tag{4}$$

is exact and $\Omega_{B/k}$ is locally free, i.e., a projective B-module. Since $\Omega_{A/k}$ is a free A-module, we conclude that X is nonsingular if and only if (4) is split exact. From the four-term exact sequence in (2.1.8)

$$0 \longrightarrow T^0(B/k, M) \longrightarrow \mathrm{Hom}(\Omega_{A/k}, M) \longrightarrow \mathrm{Hom}(I/I^2, M) \longrightarrow T^1(B/k, M) \longrightarrow 0,$$

$T^1(B/k, M) = 0$ for all $M \in \mathsf{Mod}_B$ if and only if the map

$$\mathrm{Hom}(\Omega_{A/k}, M) \longrightarrow \mathrm{Hom}(I/I^2, M)$$

is surjective for all M, which is equivalent to split exactness by considering $M = I/I^2$.

Again, by (2.1.8) we have $T^2(B/k, M) = T^2(B/A, M)$. Since X is smooth, the localizing ideal I_x at $x \in X$ is generated by a regular sequence in the regular local ring A_x. By (2.1.7), $T^2(B_x/A_x, M) = 0$ for all $M \in \mathsf{Mod}_{B_x}$. By (2.1.10), we conclude that $T^2(B/A, M) = 0$ for all $M \in \mathsf{Mod}_B$. □

Theorem 2.1.15. *A morphism of finite type $f : X \to Y$ of Noetherian schemes is smooth if and only if it is flat, and $\mathcal{T}^1(X/Y, \mathcal{F}) = 0$ for all coherent sheaves $\mathcal{F} \in \mathsf{Coh}(X)$. Furthermore, if f is smooth, then $\mathcal{T}^2(X/Y, \mathcal{F}) = 0$ for all $\mathcal{F}$.*

Proof. The problem is local, so we may assume that $X = \operatorname{Spec} B$ and $Y = \operatorname{Spec} A$ are affine and f is given by a ring homomorphism $A \to B$.

First suppose B is flat over A and $T^1(B/A, M) = 0$ for all $M \in \mathsf{Mod}_B$. Let $y \in Y$ correspond to $\mathfrak{p} \in \operatorname{Spec} A$ and let $k = k(y)$ be its residue field. Let $A' = A/\mathfrak{p}$ and $B' = B \otimes_A A' = B/\mathfrak{p}B$; then for any $M \in \mathsf{Mod}_{B'}$ we obtain $T^1(B'/A', M) = T^1(B/A, M) = 0$ by (2.1.11). Write $B' = A'[x_1, \ldots, x_n]/I$; then $T^1(B'/A', I/I^2) = 0$.

Now consider the flat base extension from A' to $\overline{k} = \overline{\mathrm{Frac}(A')}$. By (2.1.11), we have

$$T^1(B' \otimes_{A'} \overline{k}/\overline{k}, (I/I^2) \otimes \overline{k}) = 0.$$

Since $\overline{k}$ is flat over A', we have $B' \otimes \overline{k} = \overline{k}[x_1, \ldots, x_n]/\overline{I}$ and $(I/I^2) \otimes \overline{k} = \overline{I}/\overline{I}^2$, where $\overline{I} = I \otimes \overline{k}$. Then the proof of (2.1.14) shows that $\operatorname{Spec} B' \otimes \overline{k}$ is nonsingular over $\overline{k}$. Hence, the geometric fibers of f are nonsingular, and thus f is smooth.

Conversely, suppose that B is smooth over A. We claim that $T^1(B/A, B/\mathfrak{m}) = 0$ for all maximal ideals $\mathfrak{m}$ of B. Indeed, let $\mathfrak{m}$ correspond to the point $x \in \operatorname{Spec} B$, let $f(x) = y$, and let $k = k(y)$. Apply base change on $B/\mathfrak{m} \in \mathsf{Mod}_{B \otimes_A k}$, we obtain

$$T^1(B/A, B/\mathfrak{m}) \otimes_k \overline{k} = T^1(B \otimes_A k/k, B/\mathfrak{m}) \otimes_k \overline{k} = T^1(B \otimes_A \overline{k}/\overline{k}, (B/\mathfrak{m}) \otimes_k \overline{k}) = 0$$

since the geometric fibers are nonsingular. Since $k \to \overline{k}$ is faithfully flat, we conclude that $T^1(B/A, B/\mathfrak{m}) = 0$. □

2.2 Deformations of schemes and obstruction theory

Let k be a fixed algebraically closed ground field, and let C be the category of local artinian k-algebras with residue field k. Let X_0 be a scheme over k and take a $C \in \mathsf{C}$.

A **deformation** of X_0 over C is a pair (X, i), where X is a scheme flat over C, and $i : X_0 \to X$ is a closed immersion such that the induced map $i \otimes k : X_0 \to X \otimes_C k$ is an isomorphism.

Let $X_0 \hookrightarrow X$ be a deformation of X_0 over $C \in \mathsf{C}$. Suppose C' is another artinian k-algebra with residue field k, equipped with a surjective map $C' \to C$ with the kernel J annihilated by $\mathfrak{m}_{C'}$.

An **extension** of X over C' is a deformation X' of X_0 over C', together with a closed immersion $X \hookrightarrow X'$ inducing an isomorphism $X \xrightarrow{\sim} X' \times_{C'} C$. Two such extension X' and X'' are **equivalent** if there is an isomorphism of deformations $X' \xrightarrow{\sim} X''$ compatible with the respective closed immersion of X into X' and X''. Let $\mathrm{Def}(X/C, C')$ be the set of equivalence classes of such extensions of X' over C'. In particular, define two deformations of X_0 over C are equivalent if they are in the same class in $\mathrm{Def}(X_0/k, C)$.

Theorem 2.2.1. *In the situation described as above, with the additional structure as follows: let $Y_0 \subseteq X_0$ and $Y \subseteq X$ be closed subschemes such that $Y_0 = Y \times_C k$ under the identification $X_0 \xrightarrow{\sim} X \times_C k$. Fixed an extension X' of X over C'. An **(embedding) extension** of Y over C' in X' is a closed subscheme $Y' \subseteq X'$, flat over C', such that $Y' \times_{C'} C = Y$. Then the set of such extension of Y over C' in X' form a $H^0(Y_0, \mathcal{N}_{Y_0/X_0} \otimes_k J)$-pseudotorsor.*

Proof. First we consider the affine case $X = \operatorname{Spec} B$, $X' = \operatorname{Spec} B'$, $Y = \operatorname{Spec} A$, and $A = B/I$. Suppose that $Y' = \operatorname{Spec} A'$ is given by $A' = B'/I'$. Tensoring the short exact sequence

$$0 \longrightarrow I' \longrightarrow B' \longrightarrow A' \longrightarrow 0$$

with the sequence $0 \to J \to C' \to C \to 0$, we will get a diagram

$$\begin{array}{ccccccccc}
 & & 0 & & 0 & & 0 & & \\
 & & \downarrow & & \downarrow & & \downarrow & & \\
0 & \longrightarrow & J \otimes_C I & \longrightarrow & I' & \longrightarrow & I & \longrightarrow & 0 \\
 & & \downarrow & & \downarrow & & \downarrow & & \\
0 & \longrightarrow & J \otimes_C B & \longrightarrow & B' & \longrightarrow & B & \longrightarrow & 0 \\
 & & \downarrow & & \downarrow & & \downarrow & & \\
0 & \longrightarrow & J \otimes_C A & \longrightarrow & A' & \longrightarrow & A & \longrightarrow & 0 \\
 & & \downarrow & & \downarrow & & \downarrow & & \\
 & & 0 & & 0 & & 0 & &
\end{array}$$

By (2.2.4), the exactness of the bottom two rows is equivalent to the flatness of B' and A' over C, respectively. The exactness of the first column follows from the flatness of A over C. Now we seek to classify the possible A' (resp. I') full in the diagram.

Suppose that I' and I'' are two choices of I' to fill in the diagram. Given $x \in I$, lift it to $x' \in I'$ and $x'' \in I''$. Then $x'' - x' \in B'$ maps to 0 in B, and thus $x'' - x' \in J \otimes_C B$. Denote its image in $J \otimes_C A$ by $\varphi(x)$. Notice that the choice of x' and x'' may differ by elements in $J \otimes_C I$, which maps to 0 in $J \otimes_C A$. So φ is well-defined additive map. In fact, it is an B-linear homomorphism $\varphi \in \mathrm{Hom}_B(I, J \otimes_C A)$.

Conversely, given I' and $\varphi \in \mathrm{Hom}_B(I, J \otimes_C A)$, we can define another ideal I'' solving the extension problem as follows. I'' is the set of $x'' \in B'$ whose image in B is $x \in I$, and such that for any lifting $x' \in I'$ of x, the image $x'' - x$ in $J \otimes_C A$ is equal to $\varphi(x)$.

If I', I'', I''' are three choice of solution, and if φ_1 is defined by I', I'' as above, φ_2 defined by I'', I''', and φ_3 defined by I', I''', then $\varphi_3 = \varphi_1 + \varphi_2$. So we conclude that

$$\begin{aligned}\mathrm{Hom}_B(I, J \otimes_C A) \times \{\text{deformation of } Y \text{ over } C' \text{ in } X'\} &\longrightarrow \{\text{deformation of } Y \text{ over } C' \text{ in } X'\} \\ (I', \varphi) &\longmapsto I''\end{aligned}$$

is an proper transitive group action. Since we can't guarantee the existence, it only a pseudotorsor. In this case,

$$\mathrm{Hom}_B(I, J \otimes_C A) = \mathrm{Hom}_B(I, J \otimes_k A_0).$$

Since J is a k-vector space, this term becomes to

$$\mathrm{Hom}_{B_0}(I_0, J \otimes_k A_0) = \mathrm{Hom}_{A_0}(I_0 \otimes_{B_0} A_0, A_0 \otimes_k J) = H^0(Y_0, \mathcal{N}_{Y_0/X_0} \otimes_k J).$$

In general case, the group action $H^0(Y_0, \mathcal{N}_{Y_0/X_0} \otimes_k J)$ on the set of Y' can be defined locally and functorial. So we may glue it to the global action. □

Now back to deformations of schemes (rather than closed subschemes). Consider the affine scheme case first. Given a finite generated k-algebra B_0, and a deformation B of B_0 over C. We want to describe the set $\mathrm{Def}(B/C, C') := \mathrm{Def}(\mathrm{Spec}\, B/C, C')$.

Theorem 2.2.2. *In the situation as beginning:*

*(1) There is an element $\delta \in T^2(B_0/k, B_0 \otimes J)$, called the **obstruction**, with the property that $\delta = 0$ iff an extension B' of B exists.*

(2) If extension exists, then $\mathrm{Def}(B/C, C')$ form a $T^1(B_0/k, B_0 \otimes J)$-torsor.

Proof. Take a presentation $R = C[x_1, \ldots, x_n] \to B$ with kernel $I = (f_1, \ldots, f_r)$. Define a morphism $F := R^r \to I$ by $e_i \mapsto f_i$ and denoted the kernel by Q. Lift each f_i to an element $f_i' \in R' := C'[x_1, \ldots, x_n]$, and define $I' = (f_1', \ldots, f_r')$, $B' = R'/I'$, $F' = R'^r$. Now consider

$$\begin{aligned}0 \longrightarrow I' \longrightarrow R' \longrightarrow B' \longrightarrow 0 \qquad 0 \longrightarrow Q' \longrightarrow &F' \longrightarrow I' \longrightarrow 0. \\ &e_i' \longmapsto f_i'\end{aligned}$$

Tensoring C over C', we have

$$B' \otimes_{C'} C \cong \mathrm{coker}(I' \otimes_{C'} C \to R' \otimes_{C'} C) = B.$$

Tensoring with $0 \to J \to C' \to C \to 0$ over C', we get a diagram

$$\begin{array}{ccccccccc}
 & & & & 0 & & 0 & & \\
 & & & & \downarrow & & \downarrow & & \\
 & & & & Q' & \longrightarrow & Q & & \\
 & & & & \downarrow & & \downarrow & & \\
0 & \longrightarrow & F \otimes_C J & \longrightarrow & F' & \longrightarrow & F & \longrightarrow & 0 \\
 & & \downarrow & & \downarrow{\scriptstyle f'} & & \downarrow{\scriptstyle f} & & \\
0 & \longrightarrow & R_0 \otimes_k J & \longrightarrow & R' & \longrightarrow & R & \longrightarrow & 0 \\
 & & \downarrow & & \downarrow & & \downarrow & & \\
 & & B_0 \otimes_k J & \longrightarrow & B' & \longrightarrow & B & \longrightarrow & 0 \\
 & & \downarrow & & \downarrow & & \downarrow & & \\
 & & 0 & & 0 & & 0 & &
\end{array}$$

where we use $-\otimes_{C'} J = -\otimes_{C'} k \otimes_k J$ and $R_0 = k[x_1, \ldots, x_n]$. By snake lemma, there is a map $\delta_0 : Q \to B_0 \otimes_k J$ depending on the lifting f' of f. By (2.2.4), B' is flat over C' iff $\delta_0 = 0$.

Recall that in the definition of cotangent complex, any element in F_0 is of the form $f_j e_i - f_i e_j \in F$, and lifts to $f'_j e'_i - f'_i e'_j \in Q'$. So the map δ_0 factors through $\delta_1 : Q/F_0 \to B_0 \otimes_k J$. By definition of T^i functor and base change (2.1.11), we get

$$\begin{aligned}
\operatorname{coker}(\operatorname{Hom}(F/IF, B_0 \otimes_k J) \longrightarrow \operatorname{Hom}(Q/F_0, B_0 \times_k J)) = T^2(B/C, B_0 \otimes_k J) = T^2(B_0/k, B_0 \otimes_k J) \\
\bar{\delta}_1 \longmapsto \delta.
\end{aligned}$$

Here $\delta \in T^2(B_0/k, B_0 \otimes_k J)$.

We claim that δ is independent of all choices made. If we make a different choice of lifting f''_i of the f'_i, then $f'_i - f''_i$ define a map from F' to $R_0 \otimes_k J$, and hence from $F/IF \to B_0 \otimes_k J$, and these go to zero in T^2. Suppose we choose a different polynomial ring $R^* \to B$. As in the proof of (2.1.1), we may reduce to the case $R^* = R[y_1, \ldots, y_s]$, and $y_i \mapsto 0 \in B$. Then the $\delta_0^*(y_i) = 0$ implies $\delta = \delta^*$ in $T^2(B_0/k, B_0 \otimes_k J)$.

If the extension B'/C' exists, take a polynomial ring R' over C' maps surjectively to B'. Consider

$$0 \longrightarrow I' \longrightarrow R' \longrightarrow B' \longrightarrow 0 \qquad 0 \longrightarrow Q' \longrightarrow F' \longrightarrow I' \longrightarrow 0. \tag{5}$$

Since B' and R' are flat over C', we also have I' is flat over C'. Let $R = R' \otimes_{C'} C$, $I = I' \otimes_{C'} C$, and $Q = Q' \otimes_{C'} C$. Then $(5) \otimes_{C'} C$ will give a presentation to calculate cotangent complex of B over C. Since $Q' \to Q$ is surjective, it follows that $\delta_0 = 0$, and thus $\delta = 0$.

Conversely, suppose that $\delta = 0$. We claim that for a suitable choose of lifting f'_i, the induced map $\delta_0 : Q \to B_0 \otimes_k J$ is zero. By assumption, the map $\delta_1 \in \operatorname{Hom}(Q/F_0, B_0 \otimes_k J)$ lifts to a map $\gamma : F/IF \to B_0 \otimes_k J$, and defines a map $F \to B_0 \otimes_k J$. Since F is free, it lifts to a map $F \to R_0 \otimes_k J$. Denote the image of e_i by $g_i \in R_0 \otimes_k J$. Now take $f''_i = f'_i - g_i$, then the new δ_0 is zero, and thus the new B' is flat over C'. This complete the proof of *(1)*.

Suppose one such extension B_1' exists. Take a presentation $0 \to I_0 \to R_0 \to B_0 \to 0$ as usual. Lift the surjection $R_0 \to B_0$ to a map $R \to B$, and to a map $R' \to B_1'$, which is compatible with $B_1' \to B \to B_0$. By Nakayama's lemma, $R \to B$ and $R' \to B_1'$ are surjective.

For any other extension B_2' of B, the map $R \to B$ lifts to a map $R' \to B_2'$. Thus, every abstract deformation is also an embedded deformation $X \hookrightarrow \mathbb{A}_k^n$. By (2.2.1), the embedded deformation is a $\mathrm{Hom}_B(I/I^2, B \otimes_C J)$-torsor. Suppose that we have two embedding deformations B_2' and B_3' are equivalent as abstract extensions of B. Choose an isomorphism $B_2' \cong B_3'$ as extension of B over C'. Then we obtain two maps $R' \to B_2'$, which can be regarded as the lifting of $R' \to R \to B$. By (2.1.13), the differ of this two map is a C'-derivation of R' to $B_2' \otimes_{C'} J \cong B \otimes_C J$, which can be regarded as an element of

$$\mathrm{Hom}_{R'}(\Omega_{R'/C'}, B \otimes_C J) = \mathrm{Hom}_R(\Omega_{R'/C'} \otimes_{C'} C, B \otimes_C J) = \mathrm{Hom}_R(\Omega_{R/C}, B \otimes_C J).$$

Using the exact sequence (2.1.8)

$$\mathrm{Hom}_R(\Omega_{R/C}, B \otimes_C J) \longrightarrow \mathrm{Hom}(I/I^2, B \otimes_C J) \longrightarrow T^1(B/C, B \otimes_C J) \longrightarrow 0,$$

we observe that the ambiguity of embedding is exactly resolved by the image of the derivations. We conclude that $\mathrm{Def}(X/C, C')$ is a torsor under $T^1(B/C, B_0 \otimes_k J) = T^1(B_0/k, B_0 \otimes_k J)$ by (2.1.11). □

Remark 2.2.3. Given an extension B' of B over C', the automorphism group of B' as an extension of B over C is naturally isomorphic to the group $T^0(B_0/k, J \otimes_k B_0)$. Indeed, take $k = C'$ and $R = B'$ in (2.1.13), we have isomorphism

$$\begin{aligned} \mathrm{Der}_{C'}(B', J \otimes_{C'} B') &\xrightarrow{\sim} \mathrm{Aut}(B'/B) = \mathrm{Der}_{C'}(B', B) \\ \theta &\longmapsto \mathrm{id}_{B'} + \theta \end{aligned}$$

where $\theta_1 + \theta_2 \mapsto (\mathrm{id}_{B'} + \theta_1)(\mathrm{id}_{B'} + \theta_2) = \mathrm{id}_{B'} + \theta_1 + \theta_2$ since $J \otimes_{C'} B'$ is square zero ideal. Moreover, we have

$$\begin{aligned} \mathrm{Der}_{C'}(B', J \otimes_{C'} B') &= \mathrm{Hom}_{B'}(\Omega_{B'/C'}, J \otimes_{C'} B') = \mathrm{Hom}_{B_0}(\Omega_{B'/C'} \otimes_{B'} B_0, J \otimes_k B_0) \\ &= \mathrm{Hom}_{B_0}(\Omega_{B_0/k}, J \otimes_k B_0) = T^0(B_0/k, B_0 \otimes_k J). \end{aligned}$$

For general case, we recall the notation $\mathcal{T}_{X_0}^i = \mathcal{T}^i(X_0/k, \mathcal{O}_{X_0})$.

Lemma 2.2.4 (local criterion of flatness)**.** *Let $A' \to A$ be a surjective homomorphism of noetherian rings whose kernel J is square zero. Then an A'-module M' is flat over A' iff*

(1) $M = M' \otimes_{A'} A$ is flat over A

(2) the natural map $M \otimes_A J \to M'$ is injective.

Proof. Since $J^2 = 0$, we may regard J as A-module and identify $M' \otimes_{A'} J$ with $M \otimes_A J$.

If M' is flat over A', then *(1)* follows by base extension and *(2)* follows by tensoring M' with the exact sequence

$$0 \longrightarrow J \longrightarrow A' \longrightarrow A \longrightarrow 0.$$

Conversely, suppose that M' satisfies *(1)* and *(2)*. It suffices to prove $\mathrm{Tor}_1^{A'}(M', A'/\mathfrak{p}') = 0$ for every $\mathfrak{p}' \subseteq A'$. Since J is nilpotent, it is contained in $\mathfrak{p}'$. Let $\mathfrak{p} = \mathfrak{p}'/J \in \operatorname{Spec} A$, then we have the following diagram

$$\begin{array}{ccccccccc}
 & & & & 0 & & 0 & & \\
 & & & & \downarrow & & \downarrow & & \\
0 & \longrightarrow & J & \longrightarrow & \mathfrak{p}' & \longrightarrow & \mathfrak{p} & \longrightarrow & 0 \\
 & & \| & & \downarrow & & \downarrow & & \\
0 & \longrightarrow & J & \longrightarrow & A' & \longrightarrow & A & \longrightarrow & 0 \\
 & & & & \downarrow & & \downarrow & & \\
 & & & & A'/\mathfrak{p}' & & A/\mathfrak{p} & & \\
 & & & & \downarrow & & \downarrow & & \\
 & & & & 0 & & 0 & &
\end{array}$$

Tensoring with M', we obtain

$$\begin{array}{ccccccccc}
 & & & & 0 & & 0 & & \\
 & & & & \downarrow & & \downarrow & & \\
 & & & & \mathrm{Tor}_1^{A'}(M', A'/\mathfrak{p}') & \longrightarrow & \mathrm{Tor}_1^{A}(M, A/\mathfrak{p}) & & \\
 & & & & \downarrow & & \downarrow & & \\
M \otimes_A J & \longrightarrow & M' \otimes_{A'} \mathfrak{p}' & \longrightarrow & M \otimes_A \mathfrak{p} & \longrightarrow & 0 \\
\| & & \downarrow & & \downarrow & & \\
M \otimes_A J & \longrightarrow & M' & \longrightarrow & M & \longrightarrow & 0 \\
 & & \downarrow & & \downarrow & & \\
 & & M' \otimes_{A'} A'/\mathfrak{p}' & & M \otimes_A A/\mathfrak{p} & & \\
 & & \downarrow & & \downarrow & & \\
 & & 0 & & 0 & &
\end{array}$$

By *(2)*, the second (and thus also the first) horizontal sequence is exact on the left. By snake lemma, $\mathrm{Tor}_1^{A'}(M', A'/\mathfrak{p}') \cong \mathrm{Tor}_1^{A}(M, A/\mathfrak{p}) = 0$ by *(1)*. □

Theorem 2.2.5. *In the situation as beginning:*

(1) There are three successive obstructions for the existence of an extension X' of X over C':

- $\delta_1 \in H^0(X_0, \mathcal{T}^2_{X_0} \otimes J)$ *for the existence of local extension.*
- $\delta_2 \in H^1(X_0, \mathcal{T}^1_{X_0} \otimes J)$ *for the compatible isomorphism type of local extensions.*
- $\delta_3 \in H^2(X_0, \mathcal{T}^0_{X_0} \otimes J)$ *for the obstruction of gluing.*

(2) Fixed an extension X_1' of X over C', there is an exact sequence

$$0 \longrightarrow H^1(X_0, \mathcal{T}^0_{X_0} \otimes J) \longrightarrow \mathrm{Def}(X/C, C') \longrightarrow H^0(X_0, \mathcal{T}^1_{X_0} \otimes J) \longrightarrow H^2(X_0, \mathcal{T}^0_{X_0} \otimes J).$$

(3) The automorphism group of X_1' as an extension of X over C' is naturally isomorphic to the group $H^0(X_0, \mathcal{T}_{X_0} \otimes J)$.

Proof. For each open affine subscheme $U_i \subseteq X$, by (2.2.2*(1)*), there is an obstruction lying in $H^0(U_i, \mathcal{T}^2_{U_i} \otimes J)$ for the existence of an extension U_i' of U_i over C'. Since the definition of obstruction in (2.2.2*(1)*) is compatible with localization, we may glue the data to a global obstruction $\delta_1 \in H^0(X_0, \mathcal{T}^2_{X_0} \otimes J)$.

If $\delta_1 = 0$, then for each U_i, there exists an extension U_i' of U_i over C'. For each $U_{ij} = U_i \cap U_j$, we have two deformations $U_i'|_{U_{ij}}$ and $U_j'|_{U_{ij}}$. By (2.2.2*(2)*), their difference gives an element in $H^0(U_{ij}, \mathcal{T}^1 \otimes J)$. The difference of three of these is zero on U_{ijk}, so we get the second obstruction $\delta_2 \in H^1(X_0, \mathcal{T}^1_{X_0} \otimes J)$.

If $\delta_2 = 0$, then there exists isomorphism $\varphi_{ij} : U_i'|_{U_{ij}} \xrightarrow{\sim} U_j'|_{U_{ij}}$ as the extension of U_{ij} over C' for each i, j. Then $\varphi_{ij} \circ \varphi_{jk} \circ \varphi_{ki}|_{U_{ijk}}$ gives an automorphism of $U_i'|_{U_{ijk}}$. By (2.2.3), this gives an element in $H^0(U_{ijk}, \mathcal{T}^0 \otimes J)$, and hence glue to the third obstruction $\delta_3 \in H^2(X_0, \mathcal{T}^0_{X_0} \otimes J)$.

If $\delta_3 = 0$, then we can glue the extensions U_i' to get a global extension X' by gluing lemma.

For *(2)*, fixed an extension X_1' of X over C'. If X_2' is another extension, by (2.2.2), their difference on each open affine subscheme U_i gives an element of $H^0(U_i, \mathcal{T}^1_{X_0} \otimes J)$. These glue to a global element of $H^0(X_0, \mathcal{T}^1_{X_0} \otimes J)$. Conversely, a global element of $H^0(X_0, \mathcal{T}^1_{X_0} \otimes J)$ will give extensions of U_i that are isomorphic on the intersection U_{ij}, and will there is an obstruction in $H^2(X_0, \mathcal{T}^0_{X_0} \otimes J)$ to glue these to a global extension.

Suppose that two extensions X_2' and X_3' give the same element in $H^0(X_0, \mathcal{T}^1_{X_0} \otimes J)$. Then there is an isomorphism $\phi_i : X_2'|_{U_i} \xrightarrow{\sim} X_3'|_{U_i}$ as extension of U_i. Then $\psi_{ij} := \varphi_j^{-1} \circ \varphi_i$ is an automorphism of $X_2'|_{U_{ij}}$, which defines a element in $H^0(U_{ij}, \mathcal{T}^0_{X_0} \otimes J)$. These elements agree on U_{ijk}, so we get an element of $H^1(X_0, \mathcal{T}^0_{X_0} \otimes J)$. The vanishing of this element is equivalent to the existence of automorphism $\theta_i : X_2'|_{U_i} \to X_2'|_{U_i}$ as the extension of U_i such that $\psi_{ij} = \theta_j^{-1} \circ \theta_i$. Then $\varphi_i \circ \theta_i$ glue to an isomorphism between X_2' and X_3' as the extension of X. Hence we get the desired exact sequence for $\mathrm{Def}(X/C, C')$.

For *(3)*, the automorphism group of $X_1'|_{\mathrm{Spec}\, B}$ is naturally isomorphic to

$$T^0(B_0/k, B_0 \otimes J) = \mathrm{Hom}_k(\Omega_{B_0/k}, B_0 \otimes J) = T_{B_0/k} \otimes J = (\mathcal{T}^0_{X_0} \otimes J)(\mathrm{Spec}\, B).$$

Hence, the automorphism group of X_1' is naturally isomorphic to $H^0(X_0, \mathcal{T}^0_{X_0} \otimes J)$ by gluing. □

Remark 2.2.6. The assumption $\mathfrak{m}_{C'} J = 0$ is too strong for the discussion in the future. In fact, in most situations we only assume that $J^2 = 0$. The statements of the theorems above can be modified accordingly as follows:

(1) In the proof of (2.2.1), we show that the embedding extension of Y over C' in X' form a

$$\mathrm{Hom}_B(I, J \otimes_C A) = \mathrm{Hom}_A(I/I^2, J \otimes_C A) = H^0(\mathrm{Spec}\, A, \mathcal{N}_{Y/X} \otimes f^* \tilde{J})$$

pseudotorsor, where $f : Y \to \operatorname{Spec} C$ is the structure map.

(2) In the proof of (2.2.2), the obstruction live in

$$T^2(B/C, B_0 \otimes_k J) = T^2(B/C, B \otimes_C J) = H^0(\operatorname{Spec} B, \mathcal{T}^2(\operatorname{Spec} B/\operatorname{Spec} C, f^*\tilde{J}))$$

and we show that $\operatorname{Def}(B/C, C')$ form a

$$T^1(B/C, B \otimes_C J) = H^0(\operatorname{Spec} B, \mathcal{T}^1(\operatorname{Spec} B/\operatorname{Spec} C, f^*\tilde{J}))$$

pseudotorsor, where $f : \operatorname{Spec} B \to \operatorname{Spec} C$ is the structure map.

(3) Finally, (2.2.5) may be reformulated by replacing each cohomology group of the form $H^q(X_0, \mathcal{T}^p_{X_0} \otimes J)$ with

$$H^q(X, \mathcal{T}^p(X/\operatorname{Spec} C, f^*\tilde{J})),$$

where $f : X \to \operatorname{Spec} C$ is the structure map. Or more concisely, we may deonte it by $H^q(X, \mathcal{T}^p(X/C, f^*\tilde{J}))$.

Corollary 2.2.7. *If X_0 is nonsingular, then*

- *There is just one obstruction in $H^2(X_0, \mathcal{T}_{X_0} \otimes J)$ for the existence of an extension X' of X over C'.*
- *If such extension exist, their equivalence classes form a $H^1(X_0, \mathcal{T}_{X_0} \otimes J)$-torsor.*

Proof. By (2.1.14), the sheaves $\mathcal{T}^1_{X_0}$ and $\mathcal{T}^2_{X_0}$ are zero. Then the assertion follows from (2.2.5). □

2.3 Functors of Artin rings

Now we begin to formalize the idea of studying local deformations of a fixed object.

Still let k be a fixed algebraically closed ground field, and let C be the category of local artinian k-algebras with residue field k. Let R be a complete k-algebra, and for each $A \in \mathsf{C}$, let $h_R(A)$ be the set of k-algebra homomorphisms $\operatorname{Hom}(R, A)$. For any morphism $A \to B$ in C we get a map of sets $h_R(A) \to h_R(B)$, so h_R is a covariant functor from C to Set.

Definition 2.3.1. A covariant functor $F : \mathsf{C} \to \mathsf{Set}$ that is (naturally) isomorphic to a functor of the form h_R for some complete local k-algebra R is called **pro-presentable**.

To explain the nature of an isomorphism between h_R and F, let us consider more generally any homomorphism of functors $\varphi : h_R \to F$ for a complete local k-algebra R with maximal ideal $\mathfrak{m}$. In particular, for each n this will give a map $\varphi_n : \operatorname{Hom}(R, R/\mathfrak{m}^n) \to F(R/\mathfrak{m}^n)$, and the image of the quotient map of R to $R/\mathfrak{m}^n$ gives an element $\xi_n \in F(R/\mathfrak{m}^n)$. These elements ξ_n are compatible, in the sense that the natural map $R/\mathfrak{m}^{n+1} \to R/\mathfrak{m}^n$ induces a map of sets $F(R/\mathfrak{m}^{n+1}) \to F(R/\mathfrak{m}^n)$ that sends ξ_{n+1} to ξ_n. Thus the collection $\{\xi_n\}$ defines an element $\xi \in \varprojlim F(R/\mathfrak{m}^n)$. We will call such a collection $\xi = \{\xi_n\}$ a **formal family** of F over the ring R.

Here it is useful to introduce the category $\hat{\mathsf{C}}$ of complete local k-algebras with residue field k. The category $\hat{\mathsf{C}}$ contains the category C, and we can extend any functor F on C to a fucntor $\hat{F}$ from $\hat{\mathsf{C}}$ to Set by defining $\hat{F}(R) = \varprojlim F(R/\mathfrak{m}^n)$ for any $R \in \hat{\mathsf{C}}$. In this notation, $\hat{F}(R)$ is the set of formal families of F over R.

Coversely, a formal family $\xi = \{\xi_n\}$ of $\hat{F}(R)$ defines a homomorphism of functors $\varphi : h_R \to F$ as follows. For any $A \in \mathsf{C}$ and any homomorphism $f : R \to A$, since A is artinian, it factors through $R/\mathfrak{m}^n$ for some n, say $f = g\pi$, where $\pi : R \to R/\mathfrak{m}^n$ and $g : R/\mathfrak{m}^n \to A$. Then let $\varphi(f)$ be the image of ξ_n under the map $F(g) : F(R/\mathfrak{m}^n) \to F(A)$. It is easy to check that these constructions are well-defined and inverse to each other, so we have the following:

Proposition 2.3.2. *If F is a functor from C to Set and R is a complete local k-algebra with residue field k, then there is a natural bijection between the set $\hat{F}(R)$ of formal families $\{\xi_n \mid \xi_n \in F(R/\mathfrak{m}^n)\}$ and the set of homomorphisms of functors h_R to F.*

Thus, if F is pro-representable, there is an isomorphism $\xi : h_R \to F$ for some R, and we can think of ξ as an element of $\hat{F}(R)$. We say that the pair (R, ξ) **pro-repsents** the functor F. One can verify easily that if F is pro-representable, the pair (R, ξ) is unique up to isomorphism.

In many cases of interest, the functors we consider will not be pro-representable, so we define the weaker notions of having a versal family or a miniversal family.

Definition 2.3.3. Let $F : \mathsf{C} \to \mathsf{Set}$ be a functor. A pair (R, ξ) with $R \in \hat{\mathsf{C}}$ and $\xi \in \hat{F}(R)$ is a **versal family** for F if the associated map $h_R \to F$ is strongly surjective. Here we say that a morphism of functors $G \to F$ is **strongly surjective** if for every $A \in \mathsf{C}$, the map $G(A) \to F(A)$ is surjective, and furthermore, for every surjection $B \to A$ in C, the map $G(B) \to G(A) \times_{F(A)} F(B)$ is also surjective. In our case this means that given a map $R \to A$ inducing an element $\eta \in F(A)$, and given $\theta \in F(B)$ mapping to η, one can lift the map $R \to A$ to a map $R \to B$ inducing θ.

If in addition the map $h_R(D) \to F(D)$ is bijective, where $D = k[t]/t^2$ is the ring of dual numbers, we say that (R, ξ) is a **miniversal family**, or that the functor has a **pro-representable hull** (R, ξ). We say that (R, ξ) is a **universal family** if it pro-represents the functor F.

The following proposition explains the significance of this terminology.

Proposition 2.3.4. *Let (R, ξ) be a formal family of the functor F.*

(1) If (R, ξ) is a versal family, then for any other formal family (S, η), there is a ring homomorphism $f : R \to S$ such that the induced map $\hat{F}(R) \to \hat{F}(S)$ sends ξ to η.

(2) If (R, ξ) is miniversal, then for any (S, η) the map $f : R \to S$ of (a) induces a unique homomorphism $R/\mathfrak{m}_R^2 \to S/\mathfrak{m}_S^2$.

(3) If (R, ξ) is a universal family, then for any (S, η) as in (a), the corresponding map $f : R \to S$ is unique.

Proof. (1) Let (R, ξ) be a versal family, and (S, η) be any formal family. Then by definition we have a strongly surjective morphism of functor $\varphi : h_R \to F$, determined by ξ. For each n, we have an element $\eta_n \in F(S/\mathfrak{m}_S^n)$. Since φ is surjective, we can lift it to an element $\theta_n \in h_R(S/\mathfrak{m}_S^n)$, i.e., a homomorphism $\theta_n : R \to S/\mathfrak{m}_S^n$ such that the induced map $F(R/\mathfrak{m}_R^n) \to F(S/\mathfrak{m}_S^n)$ sends ξ_n to η_n. Furthermore, because of the strong surjectivity of φ, starting from θ_1, we can lift successively and obtain a compatible family $\{\theta_n\}$, that is, so that each θ_n is obtained by θ_{n+1} followed by the natural map $S/\mathfrak{m}_S^{n+1} \to S/\mathfrak{m}_S^n$. Thus the maps θ_n determine a homomorphism $R \to \varprojlim S/\mathfrak{m}_S^n$, which is equal to S, since S is a complete local ring. This map $f : R \to S$ then induces the map $\hat{F}(R) \to \hat{F}(S)$ that sends ξ to η by construction.

(2) If (R, ξ) is miniversal, to show that the induced map $\bar{f} : R/\mathfrak{m}_R^2 \to S/\mathfrak{m}_S^2$ is unique, it is sufficient to show that for every map $S/\mathfrak{m}_S^2 \to D = k[t]/t^2$, the induced map $R/\mathfrak{m}_R^2 \to D$ is unique. But since $h_R(D) \to F(D)$ is bijective, by definition, this follows immediately.

(3) If (R, ξ) is a universal family, then $h_R \to F$ is an isomorphism, so η determines a unique map $h_S \to h_R$, and hence a unique $f : R \to S$. □

2.4 Schlessinger's criterion

In this section we will prove Schlessinger's theorem, which gives criteria for a functor of Artin rings to have a versal family, a miniversal family, or to be pro-representable.

We keep the notation of the previous subsection: k is a fixed algebraically closed field, C is the category of local artinian k-algebras, and F is a covariant functor form C to Set. Note that the category C has fibered direct products. If $A' \to A$ and $A'' \to A$ are morphisms in C, we take $A' \times_A A''$ to be the set-theoretic fibered product $\{(a', a'') \mid a' \text{ and } a'' \text{ have the same image in } A\}$. The ring operations extend naturally, giving another object of C.

It is convenient to introduce the notation t_F for $F(D)$, where D is the ring of dual numbers. We call this the **tangent space** of F. Similarly t_R denotes the tangent space of the functor h_R, which is just $\mathrm{Hom}_k(R, D)$, and is equal to the dual vector space of $\mathfrak{m}_R/\mathfrak{m}_R^2$.

A **small extension** in C is a surjective map $A' \to A$ whose kernel I is a one-dimensional k-vector space.

Proposition 2.4.1. *If F has a versal family, then:*

(1) $F(k)$ has just one element.

(2) For any morphisms $A' \to A$ and $A'' \to A$ in C, the natural map

$$F(A' \times_A A'') \to F(A') \times_{F(A)} F(A'')$$

is surjective.

If furthermore F has a miniversal family, then:

(3) For any $A \in \mathsf{C}$, considering the maps $A \to k$ and $D \to k$, the map of (b) above,

$$F(A \times_k D) \to F(A) \times_{F(k)} F(D)$$

is bijective.

(4) $F(D) = t_F$ has a natural structure of a finite-dimensional k-vector space.

(5) For any small extension $p : A' \to A$ and any element $\eta \in F(A)$, there is a transitive group action of the vector space t_F on the set $p^{-1}(\eta)$ (provided it is nonempty).

Finally, if F is pro-representable, then:

(6) The maps of (b) are all bijective and the action of (e) is bijective whenever $p^{-1}(\eta)$ is nonempty.

Proof. (1) Since $\mathrm{Hom}(R, k) \to F(k)$ is surjective, and $\mathrm{Hom}(R, k)$ has just one element, so does $F(k)$.

(2) Given elements $\eta' \in F(A')$ and $\eta'' \in F(A'')$ mapping to the same element $\eta \in F(A)$, by the strong surjective property of a versal family, there are compatible homomorphisms of R to A', A, and A'' inducing these elements. Then there is a unique map of R to the product $A' \times_A A''$ inducing the given maps of R to A' and A''. This in turn defines an element of $F(A' \times_A A'')$ that restricts to η' and η'' as required.

(3) Suppose we are given $\eta \in F(A)$ and $\xi \in F(D)$. We know from *(2)* that there are elements of $F(A \times_k D)$ lying over the pair (η, ξ). Suppose θ_1 and θ_2 are two such. Choose a homomorphism $u : R \to A$ inducing η. By the strong surjectivity we can lift u to v_1 and $v_2 : R \to A[t]/t^2$, inducing θ_1 and θ_2. Since θ_1 and θ_2 both lie over ξ, the projections of v_1 and v_2 to D both induce ξ. By the hypothesis of miniversality, $t_R \to t_F$ is bijective, so these restrictions are equal. Since v_1 and v_2 also induce the same map $u : R \to A$, we obtain $v_1 = v_2$, and hence $\theta_1 = \theta_2$.

(4) By miniversality, $t_R \to t_F$ is bijective, so we can just carry over the vector space structure on t_R to t_F. But this structure can also be recovered intrinsically, using only the functorial properties of F and *(3)* above (2.4.2).

(5) Let $A' \to A$ be a small extension with kernel $I \cong k$. Note that $A' \times_A A' \cong A' \times_k k[I]$ by sending $(x, y) \mapsto (x, x_0 + y - x)$,where $x_0 \in k$ is the residue of x mod $\mathfrak{m}$. Consider the surjective map

$$F(A' \times_A A') \to F(A') \times_{F(A)} F(A')$$

of *(2)*. Using the isomorphism above and *(3)*, we can reinterpret the left-hand side as $F(A' \times_k k[I]) \cong F(A') \times t_F$, and we get a surjective map

$$F(A') \times t_F \to F(A') \times_{F(A)} F(A')$$

that is an isomorphism on the first factor. If we take $\eta \in F(A)$ and fix $\eta' \in p^{-1}(\eta)$ then we get a surjective map

$$\{\eta'\} \times t_F \to \{\eta'\} \times p^{-1}(\eta),$$

and this gives a transitive group action of t_F on $p^{-1}(\eta)$.

(6) If F is pro-representable the proof of *(2)* shows that these maps are all bijective. It follows that the action of *(5)* is also bijective. □

Example 2.4.2. Let $F : \mathsf{C} \to \mathsf{Set}$ be a functor having properties *(1)* and *(3)* of (2.4.1). Then $t_F = F(D)$ has a natural structure of a k-vector space as follows. For each $\lambda \in k$, the ring homomorphism $\lambda : D \to D,\ t \mapsto \lambda t$ induces a map $\lambda_* : t_F \to t_F$. Taking two copies of D, say $D_1 = k[t_1]/t_1^2$ and $D_2 = k[t_2]/t_2^2$, consider the ring homomorphism $D_1 \times_k D_2 \to D,\ t_i \mapsto t,\ i = 1, 2$. This induces a map $F(D_1 \times_k D_2) \to F(D)$. Using *(1)* and *(3)* this gives a map $\mu : t_F \times t_F \to t_F$. Then taking λ_* and μ as scalar multiplication and addition makes t_F into a k-vector space.

Lemma 2.4.3 (Chevalley's theorem)**.** *Let $(A, \mathfrak{m})$ be a complete local ring. Let $\{\mathfrak{a}_n\}$ be a descending sequence of ideals, with $\cap \mathfrak{a}_n = (0)$. Then for each s, there is an $n(s)$ such that for $n \geq n(s)$, $\mathfrak{a}_n \subseteq \mathfrak{m}^s$.*

Proof. [7, Vol. II, VIII, Thm. 13, p.270]. □

Theorem 2.4.4 (Schlessinger's criterion)**.** *The functor $F : \mathsf{C} \to \mathsf{Set}$ has a miniversal family if and only if:*

(H_0) $F(k)$ has just one element.

(H_1) $F(A' \times_A A'') \to F(A') \times_{F(A)} F(A'')$ is surjective for every small extension $A'' \to A$.

(H_2) The map of H_1 is bijective when $A'' = D$ and $A = k$.

(H_3) t_F is a finite-dimensional k-vector space.

Furthermore, F is pro-representable if and only if in addition:

(H_4) For every small extension $p : A'' \to A$ and every $\eta \in F(A)$ for which $p^{-1}(\eta)$ is nonempty, the group action of t_F on $p^{-1}(\eta)$ is bijective.

Proof. The necessity has been proved. So now let F be a functor satisfying conditions H_0, H_1, H_2, H_3. First we will construct a ring R and a morphism $h_R \to F$. Then we will show that it has the miniversal family property.

We will define R and the map $h_R \to F$ as an inverse limit of rings R_q and elements $\xi_q \in F(R_q)$ that we construct inductively for $q \geq 0$. We take $R_0 = k$. Note that the vector space structure on t_F is already determined by conditions H_0, H_1, H_2 (2.4.2), so that it makes sense to say in H_3 that it is finite-dimensional. Let $x_1, \ldots, x_r$ be a basis of the dual vector space t_F^*, let S be the formal power series ring $k[[x_1, \ldots, x_r]]$, with maximal ideal $\mathfrak{m}$, and take $R_1 = S/\mathfrak{m}^2$. Then $t_{R_1} \cong t_F$ by construction. Furthermore, by iterating the condition H_2, we obtain $F(R_1) = F(k[x_1] \times \cdots \times k[x_r]) \cong t_F \otimes_k t_F^*$. The natural element here gives $\xi_1 \in F(R_1)$, inducing the isomorphism $t_{R_1} \cong t_F$.

Now suppose we have constructed a compatible sequence (R_i, ξ_i) for $i = 1, \ldots, q$, with $\xi_i \in F(R_i)$, where $R_i = S/J_i$, and $\mathfrak{m}^{i+1} \subseteq J_i \subseteq J_{i-1}$, and for each i the natural map $R_i \to R_{i-1}$ sends ξ_i to ξ_{i-1}.

Then, to construct R_{q+1}, we look at ideals J in S, with $\mathfrak{m}J_q \subseteq J \subseteq J_q$, and take J_{q+1} to be the minimal such ideal J with the property that $\xi_q \in F(R_q)$ lifts to an element $\xi' \in F(S/J)$. To show that there is a minimal such J, it will be sufficient to show that if J and K are two such, then their intersection $J \cap K$ is another one. By enlarging J or K we may assume without loss of generality that $J + K = J_q$. In that case $S/(J \cap K) = (S/J) \times_{(S/J_q)} (S/K)$. Now the existence of liftings of ξ_q over S/J and S/K implies by condition H_1 the existence of a lifting over $S/J \cap K$. Then we take $R_{q+1} = S/J_{q+1}$, and ξ_{q+1} any lifting of ξ_q, which exists by construction.

Thus we obtain a surjective system of rings R_q and compatible elements $\xi_q \in F(R_q)$. Let $J = \cap J_q$ and take R to be S/J. Then R is a complete local ring and $R_q = R/\bar{J}_q$, where $\bar{J}_q = J_q/J$. Since $J_q \supseteq \mathfrak{m}^{q+1}$ for each q, and conversely, for each s, some $\bar{J}_q \subseteq \mathfrak{m}_R^s$ (2.4.3), it follows that the ideals $\bar{J}_q$ form a base for the $\mathfrak{m}_R$-adic topology of R. Therefore $R = \varprojlim R_q$, and we define $\xi = \varprojlim \xi_q \in \hat{F}(R)$. This gives the desired map $h_R \to F$.

I claim that (R, ξ) is a miniversal family for F. Since $t_R \cong t_F$ by construction, we have only to show for any surjective map $A' \to A$ and any $\eta' \in F(A')$ restricting to $\eta \in F(A)$, and any map $R \to A$ inducing η, that there exists a lifting to a map $R \to A'$ inducing η'. Since any surjective map factors into a sequence of small extensions, it suffices to treat the case of a small extension $A' \to A$.

Let $u : R \to A$ induce η. It will be sufficient to show that u lifts to some map $v : R \to A'$. For then v will induce an element $\eta'' \in F(A')$ lying over η. Because of 5, whose proof used only 3, which is our H_2, there is an element of t_F sending η'' to η' by the group action. This same $t_F = t_R$ acts on the set of $v : R \to A'$ restricting to u, so then we can adjust v to a homomorphism $v' : R \to A'$ inducing η'.

Thus it remains to show that for a small extension $A' \to A$, the given map $u : R \to A$ lifts to a map $v : R \to A'$. Since A is an Artin ring, u factors through R_q for some q. On the other hand, R is a quotient of the power series ring S, and the map u will lift to a map of S into A'. Thus we get a commutative diagram

$$\begin{array}{ccccc} S & \xrightarrow{w} & R_q \times_A A' & \longrightarrow & A' \\ \downarrow & & \downarrow{\scriptstyle p'} & & \downarrow{\scriptstyle p} \\ R & \longrightarrow & R_q & \longrightarrow & A \end{array}$$

Note that $p' : R_q \times_A A' \to R_q$ is also a small extension. If this map has a section $s : R_q \to R_q \times_A A'$, then using s and the second projection we get a map $R_q \to A'$ lifting u, and we are done.

If p' does not have a section, then I claim that the map w is surjective. Indeed, if w is not surjective, then $\operatorname{Im} w$ is a subring mapping surjectively to R_q. The kernel of $\operatorname{Im} w \to R_q$ must be strictly contained in $I = \ker p'$, which is a one-dimensional vector space, so this kernel is zero, the map $\operatorname{Im} w \to R_q$ is an isomorphism, and this gives a section. Contradiction!

Thus knowing that w is surjective, let $J = \ker w$. Then $J \subseteq J_q$, since S maps to R_q via w. On the other hand, $J \supseteq \mathfrak{m}J_q$, since p' is a small extension. But also $\xi_q \in F(R_q)$ and there is an $\eta' \in F(A')$ lying over $\eta \in F(A)$, so by H_1, there is a $\xi' \in F(R_q \times_A A')$ lying over both of these. Since $R_q \times_A A' = S/J$, this ideal J satisfies the condition imposed in the construction. Therefore $J \supseteq J_{q+1}$ and w factors through R_{q+1}. This

gives the required lifting of R to A', and completes the proof that (R, ξ) is a miniversal family for F.

Finally, suppose in addition that F satisfies H_4. To show that $h_R(A) \to F(A)$ is bijective for all A, it will be sufficient to show inductively, strating $A = k$, that for any small extension $p : A' \to A$ we have that $\mathrm{Hom}(R, A') \to \mathrm{Hom}(R, A) \times_{F(A)} F(A')$ is bijective. So fix $u \in \mathrm{Hom}(R, A)$ and the corresponding $\eta \in F(A)$. If there is no map of R to A' lying over u, then there is also no $\eta' \in F(A')$ lying over η, and there is nothing to prove. On the other hand, if $p^{-1}(\eta)$ is nonempty, then the action of t_F on $p^{-1}(\eta)$ is bijective by H_4, and the action of t_R on the set of homomorphism $R \to A'$ lying over u is bijective, since h_R is pro-representable, and $t_R \cong t_F$ by miniversality, so our map is bijective as required. □

2.5 Deformation functor

We now discuss the question of pro-representability or existence of a miniversal family of the deformation functor of a scheme.

Let X_0 be a scheme over k and A be a Artin ring with residue field k. Define the following **deformation fucntor**:

$$F(A) := \mathrm{Def}(X_0/k, A)$$

Theorem 2.5.1. *Let X_0 be a scheme over k. Then the functor F has a miniversal family under either of the two folloing hypotheses:*

(1) X_0 is affine with isolated singularities and of finite type.

(2) X_0 is projective.

Proof. We verify the conditions of Schlessinger's criterion (2.4.4).

(H_0) $F(k)$ consists of the single object (X_0, id). If σ is an automorphism of X_0, the object (X_0, σ) is isomorphic to (X_0, id) by the map $\sigma : X_0 \to X_0$.

(H_1) Suppose we are given a small extension $A'' \to A$ and any map $A' \to A$, and suppose we are given objects $X' \in F(A')$, $X'' \in F(A'')$ restricting to $X \in F(A)$. Then $X' \otimes_{A'} A \cong X$, the isomorphism being compatible wtih the maps from X_0, so we can choose a closed immersion $X \hookrightarrow X'$ inducing this isomorphism. Similarly choose $X \hookrightarrow X''$. Then we define X^* by the fibered product of sheaves of rings $\mathcal{O}_{X^*}\mathcal{O}_{X'} \times_{\mathcal{O}_X} \mathcal{O}_{X''}$, and X^* will be an object of $F(A^*)$ reducing to X' and X'', where $A^* = A' \times_A A''$.

(H_2) Suppose $A = k$ in the situation of (H_1) (which effectively means $A'' \cong k[t]$) and let X^* be constructed as in (H_1). If W is any other object of $F(A^*)$ restricting to X' and X'' respectively, then we can choose immersions $X' \hookleftarrow W$ and $X'' \hookrightarrow W$ inducing these isomorphisms. Since these maps are all compatible with the immersion from X_0, they agree with the chosen maps $X \hookrightarrow X'$ and $X \hookrightarrow X''$, since in this case $X = X_0$. Now by the universal property of fibered product of rings, there is a map $X^* \to W$ compatible with the above maps. Since X^* and W are both flat over A^*, and the map becomes an isomorphism when restricted to X_0, we find that X^* is isomorphic to W, and hence they are equal as elements of $F(A^*)$.

(H_3) Here is the only place in the proof that we need the hypothesis *(1)* or *(2)*.

(1) Let $X_0 = \operatorname{Spec} B$. Then $t_F = T^1_{B/k}$ by (2.2.2). This module is supported at the finite number of singular points of X_0 (2.1.14), so has finite length (2.1.9), i.e., t_F is a finite-dimensional vector space.

(2) For arbitrary X_0, the tangent space t_F corresponds to deformations over the ring of dual numbers D. Because of the exact sequence (2.2.5*(2)*)

$$0 \longrightarrow H^1(X_0, \mathcal{T}_{X_0}) \longrightarrow \operatorname{Def}(X_0/k, D) \longrightarrow H^0(X_0, \mathcal{T}^1_{X_0}) \longrightarrow \cdots,$$

we see that if X_0 is projective, the two outside groups are finite-dimensional vector spaces, and so $\operatorname{Def}(X_0/k, D)$ is also.

Thus conditions (H_0)–(H_3) are satisfied, and F has a miniversal family. □

Next, we consider conditions under which the functor F is actually pro-representable.

Theorem 2.5.2. *Let X_0/k be given and assume the hypotheses of (2.5.1) satisfied. Then the functor F of deformations of X_0 is pro-representable if and only if for each small extension $A' \to A$, and for each deformation X' over A' restricting to a deformation X over A, the natural map $\operatorname{Aut}(X'/X_0) \to \operatorname{Aut}(X/X_0)$ of automorphisms of X' (and X) restricting the identity automorphism of X_0 is surjective.*

Proof. Suppose that $\operatorname{Aut}(X'/X_0) \to \operatorname{Aut}(X/X_0)$ is surjective for every X' lying over X. We need to verify condition (H_4) of (2.4.4). Let X'_1, X'_2 be elements of $F(A')$ inducing the same element X of $F(A)$. If $X \hookrightarrow X'_1$ and $X \hookrightarrow X'_2$ are maps inducing the isomorphisms $X'_1 \otimes_{A'} A \cong X$ and $X'_2 \otimes_{A'} A \cong X$, and if X'_1 and X'_2 are isomorphic as deformations of X_0, then I claim that the inclusions $X \hookrightarrow X'_1$ and $X \hookrightarrow X'_2$ are isomorphic as extensions of X over A'.

Indeed, let $u' : X'_1 \to X'_2$ be an isomorphism over X_0. Then $u = u' \otimes_{A'} A$ is an automorphism of X over X_0. By hypothesis this lifts to an automorphism σ of X'_1. Then $v = u \circ \sigma^{-1} : X'_1 \to X'_2$ is an isomorphism inducing the identity on X, so X'_1 and X'_2 are equivalent as extensions of X over A'.

Now by (2.2.5*(2)*), $\operatorname{Def}(X/A, A')$ is a principal homogeneous space under the action of t_F ($D \to k$ is also a small extension), so condition (H_4) of Schlessinger's criterion is satisfied, and F is pro-representable.

Conversely, suppose that F is pro-representable, let $X' \in F(A')$ restrict to $X \in F(A)$, and choose a map $u : X \hookrightarrow X'$ inducing the isomorphism $X \xrightarrow{\sim} X' \otimes_{A'} A$. Let $\sigma \in \operatorname{Aut}(X/X_0)$. Then $u' = u \circ \sigma : X \hookrightarrow X'$ gives another element of $\operatorname{Def}(X/A, A')$, and so u and u' differ by an element of t_F, by (2.2.5*(2)*). But u and u' define the same element $X' \in F(A')$, lying over X, so by condition (H_4), this element of t_F must be zero. Hence u and u' are equal as elements of $\operatorname{Def}(X/A, A')$; in other words there exists an isomorphism $\tau : X' \to X'$ over X_0 such that $u' = \tau \circ u$. Restricting to X we obtain $\sigma = \tau|_X$. Thus $\tau \in \operatorname{Aut}(X'/X_0)$ lifts σ, and tha map is surjective. □

Corollary 2.5.3. *Let X_0/k be a projective scheme with $H^0(X_0, \mathcal{T}_{X_0}) = 0$. Then the functor of deformations of X_0/k is pro-representable.*

Proof. We will show, by induction on the length of A, that for any deformation X of X_0 over A, $\mathrm{Aut}(X/X_0) = \{\mathrm{id}\}$. Then obviously the condition of (2.5.2) will be satisfied.

We start the induction by noting that $\mathrm{Aut}(X_0/X_0) = \{\mathrm{id}\}$. Inductively, assume that $\mathrm{Aut}(X/X_0) = \{\mathrm{id}\}$, where X is a deformation over A. Consider a small extension $A' \to A$ and any $X' \in F(A')$ restricting to X. Choose a map $X \hookrightarrow X'$ inducing the isomorphism $X \xrightarrow{\sim} X' \otimes_{A'} A$. Any automorphism of X' restricts to the identity on X, by the induction hypothesis, so it is an automorphism of the deformation $X \hookrightarrow X'$. Since these are classfied by $H^0(X_0, \mathcal{T}_{X_0}) = 0$ (2.2.5*(3)*), this automorphism is the identity. □

Example 2.5.4. Let X_0 be a nonsingular projective curve over the algebraically closed field k. If the genus is zero, $X_0 = \mathbb{P}^1_k$. In this case $H^0(X_0, \mathcal{T}_{X_0})$ has dimension 3, but still, it can be proved that the functor of deformation of X_0/k is pro-representable. Thus the condition of (2.5.3) is not necessary for the functor to be pro-representable.

2.6 Ran-Kawamata T^1 lifting

Definition 2.6.1. An **obstruction theory** for a functor $F : \mathsf{C} \to \mathsf{Set}$ is a k-vector space V, together with, for every exact sequence $0 \to J \to C' \to C \to 0$ such that $\mathfrak{m}_{C'} J = 0$, and for every $u \in F(C)$, an element $\varphi(u, C') \in V \otimes_k J$ such that

- $\varphi(u, C') = 0$ if and only if $u \in \mathrm{Im}\, F(C' \to C)$.
- φ is functorial in the sense that if

$$\begin{array}{ccccccccc} 0 & \longrightarrow & J_1 & \longrightarrow & C_1' & \longrightarrow & C_1 & \longrightarrow & 0 \\ & & \downarrow & & \downarrow & & \downarrow & & \\ 0 & \longrightarrow & J_2 & \longrightarrow & C_2' & \longrightarrow & C_2 & \longrightarrow & 0 \end{array}$$

  is a commutative diagram such that $\mathfrak{m}_{C_i'} J_i = 0$, then we have the following commutative diagram

$$\begin{array}{ccccc} F(C_1') & \longrightarrow & F(C_1) & \longrightarrow & V \otimes_k J_1 \\ \downarrow & & \downarrow & & \downarrow \\ F(C_2') & \longrightarrow & F(C_2) & \longrightarrow & V \otimes_k J_2. \end{array}$$

Corollary 2.6.2. *The deformation functor of smooth scheme X over k has obstruction theory.*

Proof. It follows from (2.2.7) and take $V = H^2(X, \mathcal{T}_X)$. □

Now let k be a algebraically closed field with $\mathrm{char}\, k = 0$ and $F : \mathsf{C} \to \mathsf{Set}$ still be a deformation functor. We also assume that the obstruction can be calculated by a k-vector space T^2 (2.6.1).

For $n \in \mathbb{Z}_{\geq 0}$, let

$$A_n = k[t]/(t^{n+1}),\ B_n = A_n \otimes A_1 = k[x, y]/(x^{n+1}, y^2),\ C_n = k[x, y]/(x^{n+1}, x^n y, y^2).$$

We have natural homomorphism $\alpha_n : A_{n+1} \to A_n$, $\beta_n : B_n \to A_n$, and $\gamma_n : B_n \to C_n$.

For $X_n \in F(A_n)$, let

$$T^1(X_n/A_n) = \{Y_n \in F(B_n) \mid F(\beta_n)(Y_n) = X_n\}.$$

Theorem 2.6.3 ([8]). *Assume that F is pro-representable by (R, ξ), and that the following T^1 lifting property holds: for any positive integer n and any $X_{n+1} \in F(A_{n+1})$, the natural homomorphism*

$$T^1(\alpha_n) : T^1(X_{n+1}/A_{n+1}) \to T^1(X_n/A_n)$$

is surjective for $X_n = F(\alpha_n)(X_{n+1})$. Then F is unobstructed.

Proof. It suffices to show that $F(\alpha_n)$ is surjective for all n. Indeed, according to the proof of Schlessinger's criterion, we have $R = k[[x_1, \ldots, x_r]]/J$, where $J \subseteq \mathfrak{m}_R^2$ and $r = \dim t_F$. Since $J \subseteq \mathfrak{m}_R^2$, we can define the morphism $g_2 : R \to A_1$ by $x_i \mapsto a_i t$ for any $a_i \in k$. Since $F(\alpha_n)$ is surjective for all n, there is a lifting $g_n : R \to A_n$ of g_{n-1} by induction for all $n \geq 3$. This define a morphism $g : R \to k[[t]]$ by $x_i \mapsto a_i t$. Suppose that J contains a nonzero power series $f = f_d + f_{d+1} + \cdots$, where f_n is the homogeneous polynomial of degree n, $f_d \neq 0$ and $d \neq 0$. Take a_i such that $f_d(a_1, \ldots, a_r) \neq 0$, then

$$g(f) = t^d f_d(a_1, \ldots, a_r) + t^{d+1} f_{d+1}(a_1, \ldots, a_r) + \cdots \neq 0$$

leading a contradiction. Hence, $R = k[[x_1, \ldots, x_r]]$ is a power-series ring over k, $F = h_R$ is clearly unobstructed for lifting of any map in C.

Consider the following commutative diagram

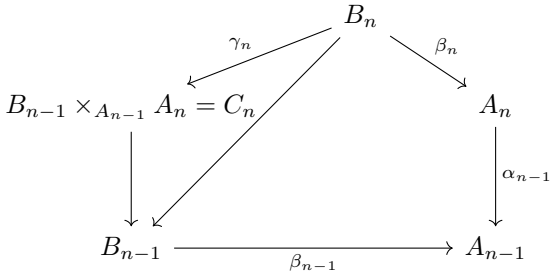
$$\begin{array}{ccccc} F(A_{n+1}) & \xrightarrow{F(\alpha_n)} & F(A_n) & \xrightarrow{\delta_1} & T^2 \otimes (t^{n+1}) \\ \downarrow{\scriptstyle F(\varepsilon_n)} & & \downarrow{\scriptstyle F(\varepsilon'_n)} & & \downarrow{\scriptstyle \varphi} \\ F(B_n) & \xrightarrow{F(\gamma_n)} & F(C_n) & \xrightarrow{\delta_2} & T^2 \otimes (x^n y) \end{array}$$

where $\varepsilon_n(t) = x + y$ and $\varepsilon'_n(t) = x + y$. By the definition of T^2, we have

$$\operatorname{Im} F(\alpha_n) = \{X_n \in F(A_n) \mid \delta_1(X_n) = 0\},\ \operatorname{Im} F(\gamma_n) = \{Y_n \in F(C_n) \mid \delta_2(Y_n) = 0\}.$$

Notice that φ is an isomorphism, since $\varphi : t^{n+1} \mapsto (n+1)x^n y$ and $\operatorname{char} k = 0$. Hence, to show that $F(\alpha_n)$ is surjective, it suffices to show that $F(\gamma_n)$ is surjective.

From the following diagram



and the pro-representability of F, we have

$$F(B_n) \xrightarrow{F(\gamma_n)} F(C_n) \cong F(B_{n-1}) \times_{F(A_{n-1})} F(A_n).$$

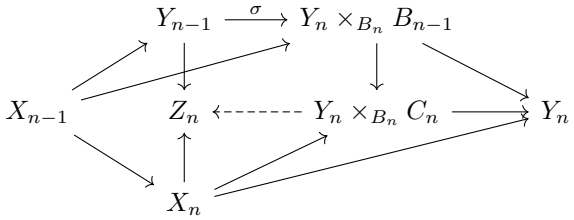
By the T^1 lifting property, for any $Z_n \in F(C_n)$, there exists $Y_n \in F(B_n)$ whose image in $F(B_{n-1})$ and $F(A_n)$ coincide with image of Z_n, respectively. Hence, $F(\gamma_n)(Y_n) = Z_n$. □

Remark 2.6.4. In [9], Kawamata replaced the definition of T^1 space by $T^1(X_n/A_n) := \mathrm{Def}(X_n/A_n, B_n)$. With this new definition, the above theorem holds without the assumption that F is pro-representable.

As in the argument above, it suffices to show that the map $F(\gamma_n) : F(B_n) \to F(C_n)$ is surjective. For $Z_n \in F(C_n)$, let

$$Y_{n-1} = Z_n \times_{C_n} B_{n-1},\ X_n = Z_n \times_{C_n} A_n,\ X_{n-1} = Z_n \times_{C_n} A_{n-1}.$$

By the T^1 lifting property, let $[Y_n]$ be the equivalent classes in $\mathrm{Def}(X_n/A_n, B_n)$ maps to $[Y_{n-1}] \in \mathrm{Def}(X_{n-1}/A_{n-1}, B_{n-1})$. Then there exists an isomorphism $\sigma : Y_{n-1} \to Y_n \times_{B_n} B_{n-1}$ compatible with the closed immersion of X_{n-1}, and we get a diagram



By (2.6.8), there exists an isomorphism $Y_n \times_{B_n} C_n \xrightarrow{\sim} Z_n$ compatible with the closed immersion of X_n. This show that $F(\gamma_n)(Y_n) = Z_n$.

Example 2.6.5. Suppose X_n is a smooth deformation of X over A_n. By (2.2.6), the deformation space can be computed via the following exact sequence

$$0 \longrightarrow H^1(X_n, \mathcal{T}^0(X_n/A_n, f^*\tilde{J})) \longrightarrow \mathrm{Def}(X_n/A_n, B_n) \longrightarrow H^0(X_n, \mathcal{T}^1(X_n/A_n, f^*\tilde{J})),$$

where $f : X_n \to A_n$ is the structure morphism and $J = \ker \beta_n$. Since $\ker \beta_n$ is a free A_n-module and X_n is smooth over A_n, it follows from (2.1.15) that

$$\mathcal{T}^i(X_n/A_n, f^*\tilde{J}) = \mathcal{T}^i_{X_n/A_n} = \begin{cases} \mathcal{T}_{X_n/A_n} & i = 0, \\ 0 & i = 1. \end{cases}$$

So we conclude that $\mathrm{Def}(X_n/A_n, B_n) = H^1(X_n, \mathcal{T}_{X_n/A_n})$.

Lemma 2.6.6. *Let A be a ring, J be a nilpotent ideal in A, and $u : M \to N$ be a homomorphism of A-modules, with N falt over A. If $\bar{u} : M/JM \to N/JN$ is an isomorphism, then u is an isomorphism.*

Proof. Let $K = \mathrm{coker}\, u$. Tensor the exact sequence

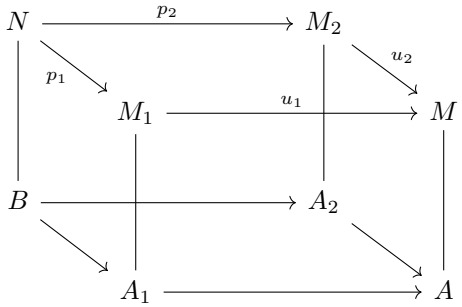
$$M \to N \to K \to 0$$

with A/J, we have $K/JK = 0$. Since J is nilpotent, we have $K = 0$. If $K' = \ker u$, then we get an exact sequence

$$0 \to K'/JK' \to M/JM \to N/JN \to 0$$

by the flatness of N. Hence, $K' = 0$. □

Lemma 2.6.7. *Consider a commutative diagram*



of compatible ring and module homomorphisms, where $B = A_1 \times_A A_2$, $N = M_1 \times_M M_2$, *and* M_i *is a flat* A_i*-module for* $i = 1, 2$. *Suppose that* A_1 *is an artinian ring and*

(1) $A_2/J \xrightarrow{\sim} A$, *where* J *is an nilpotent ideal in* A_2.

(2) u_i *induces* $M_i \otimes_{A_i} A \xrightarrow{\sim} M$.

Then N *is flat over* B, *and* p_i *induces* $N \otimes_B A_i \xrightarrow{\sim} M_i$.

Proof. Let $\{x_{1,i}\}_{i\in I} \subseteq M_1$ such that $\{\bar{x}_{1,i}\}_{i\in I}$ form a basis of $M_1/\mathfrak{m}_{A_1}M_1$. Then $\bigoplus_{i\in I} A_1 \to M_1$ defined by $e_i \mapsto x_{1,i}$ is an isomorphism by (2.6.6). Hence M_1 is a free A_1-module with basis $\{x_{1,i}\}_{i\in I}$. By *(2)*, M is the free module with basis $\{u_1(x_{1,i})\}_{i\in I}$. Choosing $x_{2,i} \in M_2$ such that $u_2(x_{2,i}) = u_1(x_{1,i})$. Then A_2-module homomorphism $\sum A_2 x_{2,i} \to M_2$ is isomorphism after modulo the ideal J. Again, by (2.6.6), M_2 is free module with basis $\{x_{2,i}\}_{i\in I}$, and it is clear that N is free B-module with basis $\{x_{1,i} \times x_{2,i}\}_{i\in I}$. Then the projection p_i induces the isomorphism $N \otimes_B A_i \xrightarrow{\sim} M_i$. □

Lemma 2.6.8. *With the notations in (2.6.7). Let* L *be a* B*-module satisfies the commutative diagram*

$$\begin{array}{ccc} L & \xrightarrow{q_2} & M_2 \\ {\scriptstyle q_1}\downarrow & & \downarrow{\scriptstyle u_2} \\ M_1 & \xrightarrow{u_1} & M \end{array}$$

where q_1 *induces* $L \otimes_B A_1 \xrightarrow{\sim} M_1$. *Then the canonical morphism* $q_1 \times q_2 : L \to N = M_1 \times_M M_2$ *is an isomorphism.*

Proof. Apply (2.6.6) to the morphism $u = q_1 \times q_2$. Note that $B/(0 \times J) \xrightarrow{\sim} A_1$. □

2.7 Deformations of smooth Calabi-Yau varieties

Before proving the unobstructedness of deformations of Calabi-Yau varieties, we need the following essential technique.

Theorem 2.7.1. *Let* $f : X \to S = \operatorname{Spec} A_n$ *be a projective and smooth morphism, where* $A_n = \mathbb{C}[t]/(t^{n+1})$. *Then* $H^q(X, \Omega^p_{X/S})$ *is a free* A_n*-module.*

Proof. Let $A = A_n$. Consider the $\mathfrak{m}_A$-adic filtration

$$\Omega^{\bullet,\mathrm{an}}_{X/S} \supseteq \mathfrak{m}_A \Omega^{\bullet,\mathrm{an}}_{X/S} \supseteq \mathfrak{m}_A^2 \Omega^{\bullet,\mathrm{an}}_{X/S} \supseteq \cdots \supseteq \mathfrak{m}_A^N \Omega^{\bullet,\mathrm{an}}_{X/S} = 0$$

of $\Omega^{\bullet,\mathrm{an}}_{X/S}$, which is biregular, and denote the associated graded object by

$$\mathrm{gr}\, \Omega^{\bullet,\mathrm{an}}_{X/S} = \bigoplus \mathrm{gr}_i \Omega^{\bullet,\mathrm{an}}_{X/S},$$

where $\mathrm{gr}_i \Omega^{\bullet,\mathrm{an}}_{X/S} := \mathfrak{m}_A^i \Omega^{\bullet,\mathrm{an}}_{X/S} \Big/ \mathfrak{m}_A^{i+1} \Omega^{\bullet,\mathrm{an}}_{X/S}$. Notice that

$$\mathrm{gr}_0 \Omega^{\bullet,\mathrm{an}}_{X/S} = \Omega^{\bullet,\mathrm{an}}_{X/S} \Big/ \mathfrak{m}_A \Omega^{\bullet,\mathrm{an}}_{X/S} = \Omega^{\bullet,\mathrm{an}}_{X_0/\mathbb{C}}$$

gives a acylic resolution of the constant sheaf $\mathbb{C}$, where X_0 is the smooth fiber of $\mathrm{Spec}\, \mathbb{C} \hookrightarrow \mathrm{Spec}\, A$. Then

$$\mathrm{gr}_{\mathfrak{m}_A} A \otimes_{\mathbb{C}} \mathrm{gr}_0 \Omega^{\bullet,\mathrm{an}}_{X/S} = \bigoplus_{n \geq 0} \mathfrak{m}_A^n / \mathfrak{m}_A^{n+1} \otimes_{A/\mathfrak{m}_A} \Omega^{\bullet,\mathrm{an}}_{X/S} \Big/ \mathfrak{m}_A \Omega^{\bullet,\mathrm{an}}_{X/S} = \bigoplus_{n \geq 0} \mathfrak{m}_A^n \Omega^{\bullet,\mathrm{an}}_{X/S} \Big/ \mathfrak{m}_A^{n+1} \Omega^{\bullet,\mathrm{an}}_{X/S} = \mathrm{gr}_{\mathfrak{m}_A} \Omega^{\bullet,\mathrm{an}}_{X/S}$$

resolves $\mathrm{gr}_{\mathfrak{m}_A} A$, and hence $\Omega^{\bullet,\mathrm{an}}_{X/S}$ gives a acylic resolution of the constant sheaf A. Since X is proper, by the GAGA principle we have

$$\mathsf{R}^n \Gamma(X, \Omega^{\bullet}_{X/S}) \cong \mathsf{R}^n \Gamma(X^{\mathrm{an}}, \Omega^{\bullet,\mathrm{an}}_{X/S}) = H^n(X^{\mathrm{an}}, A) = H^n(X^{\mathrm{an}}, \mathbb{C}) \otimes_{\mathbb{C}} A,$$

and thus

$$\mathrm{length}_A \mathsf{R}^n \Gamma(X, \Omega^{\bullet}_{X/S}) = \mathrm{length}_A(A) \cdot \dim_{\mathbb{C}} H^n(X^{\mathrm{an}}, \mathbb{C}). \tag{6}$$

Lemma 2.7.2. *Let $Y = \mathrm{Spec}\, A$ be a noetherian affine scheme, and let $f : X \to Y$ be a proper morphism. For any complex $\mathscr{F}^{\bullet} \in \mathsf{D}^+(\mathrm{Coh}(X))$ that is flat over Y, there exists a complex $\mathscr{L}^{\bullet} \in \mathsf{D}^+(\mathrm{Coh}(Y))$, whose terms $\mathscr{L}^i$ has the form $\mathcal{O}_Y^{m_i}$, and a functorial quasi-isomorphism*

$$f_*(\mathscr{F}^{\bullet} \otimes_{\mathcal{O}_X} f^* \mathscr{G}) \longrightarrow \mathscr{L}^{\bullet} \otimes_{\mathcal{O}_Y} \mathscr{G}$$

for any $\mathscr{G} \in \mathrm{Qcoh}(Y)$. Moreover, for arbitrary base change diagram

$$\begin{array}{ccc} X' & \xrightarrow{u'} & X \\ \downarrow{\scriptstyle f'} & & \downarrow{\scriptstyle f} \\ Y' & \xrightarrow{u} & Y \end{array}$$

there is a natural quasi-isomorphism

$$f'_*(u'^* \mathscr{F}^{\bullet} \otimes_{\mathcal{O}_{X'}} f'^* \mathscr{G}') \longrightarrow u^* \mathscr{L}^{\bullet} \otimes_{\mathcal{O}_{Y'}} \mathscr{G}'$$

for any $\mathscr{G}' \in \mathrm{Qcoh}(Y')$.

Proof. [10, Prop 6.10.5] □

Since f is smooth, the sheaf $\Omega_{X/S}$ is locally free on X. Applying (2.7.2) to the pair $(\mathscr{F}^\bullet, \mathscr{G}) = (\Omega^p_{X/S}, \mathcal{O}_Y)$ (Here we consider $\Omega^p_{X/S}$ as a complex concentrated in degree 0) and the base change morphism $u : \operatorname{Spec}\mathbb{C} \to \operatorname{Spec} A$, we obtain

$$\mathsf{R}f_*\Omega^p_{X/S} \cong \mathscr{L}^\bullet,\ \mathsf{R}f'_*\Omega^p_{X_0/\operatorname{Spec}\mathbb{C}} \cong \mathsf{R}f'_*(u'^*\Omega^p_{X/S}) \cong u^*\mathscr{L}^\bullet.$$

Therefore

$$H^q(X, \Omega^p_{X/S}) = H^0(\operatorname{Spec} A, \mathsf{R}^q f_*\Omega^p_{X/S}) = \Gamma(\operatorname{Spec} A, H^q(\mathscr{L}^\bullet))$$

is finite generated A-module, and the natural isomorphism of base change

$$H^q(X, \Omega^p_{X/S}) \otimes_A \mathbb{C} \cong \Gamma(\operatorname{Spec}\mathbb{C}, H^q(u^*\mathscr{L}^\bullet)) = H^q(X_0, \Omega^p_{X_0}).$$

Lemma 2.7.3. *Let $(A, \mathfrak{m}_A)$ be the noetherian local ring with residue field k. For any finite generated A-module M, we have the inequality*

$$\operatorname{length}_A(M) \leq \operatorname{length}_A(A) \cdot \dim_k(M \otimes_A k).$$

Equality holds if and only if M is free A-module.

Proof. Let $d = \dim_k(M \otimes_A k)$. By Nakayama's lemma, there exists a surjection $f : A^d \twoheadrightarrow M$. Then

$$\operatorname{length}_A(M) = d \cdot \operatorname{length}_A(A) + \operatorname{length}_A(\ker f)$$

and $\operatorname{length}_A(\ker f) = 0$ if and only if $\ker f = 0$. □

By the above lemma, we have

$$\operatorname{length}_A H^q(X, \Omega^p_{X/S}) \leq \operatorname{length}_A(A) \cdot \dim H^q(X_0, \Omega^p_{X_0}) \tag{7}$$

and equality holds if and only if $H^q(X, \Omega^p_{X/S})$ is free A-module. Consider Hodge to de Rham spectral sequence

$$E_1^{p,q} = H^q(X, \Omega^p_{X/S}) \Longrightarrow \mathsf{R}^{p+q}\Gamma(\Omega^\bullet_{X/S}).$$

Since $E_\infty^{p,q}$ is the subquotient of $E_1^{p,q}$, we deduce that

$$\operatorname{length}_A \mathsf{R}^n\Gamma(\Omega^\bullet_{X/S}) = \sum_{p+q=n} \operatorname{length}_A E_\infty^{p,q} \leq \sum_{p+q=n} \operatorname{length}_A H^q(X, \Omega^p_{X/S}). \tag{8}$$

Finally, combining (6), (7), and (8), we obtain that

$$\begin{aligned} \operatorname{length}_A(A) \cdot \dim_{\mathbb{C}} H^n(X^{\mathrm{an}}, \mathbb{C}) &= \operatorname{length}_A \mathsf{R}^n\Gamma(\Omega^\bullet_{X/S}) \\ &\leq \sum_{p+q=n} \operatorname{length}_A H^q(X, \Omega^p_{X/S}) \\ &\leq \operatorname{length}_A(A) \sum_{p+q=} \dim H^q(X_0, \Omega^p_{X_0}). \end{aligned} \tag{9}$$

By Hodge decomposition in analytic setting (X_0 is projective so X_0^{an} is Kähler) and the GAGA principle, we have

$$H^n(X^{\mathrm{an}},\mathbb{C}) = H^n(X_0^{\mathrm{an}},\mathbb{C}) \cong \bigoplus_{p+q=n} H^q(X_0^{\mathrm{an}},\Omega^{p,\mathrm{an}}_{X_0}) \cong \bigoplus_{p+q=n} H^q(X_0,\Omega^p_{X_0}),$$

and hence equality holds throughout in (9). It follows that each

$$H^q(X,\Omega^p_{X/S})$$

is a free A-module. □

Theorem 2.7.4. *Let X be a smooth projective variety over $\mathbb{C}$. If $\omega_X \cong \mathcal{O}_X$, then the flat deformations of X (cf. 2.6.2) are unobstructed.*

Proof. By (2.6.4), it suffices to show that

$$\mathrm{Def}(X_{n+1}/A_{n+1},B_{n+1}) \longrightarrow \mathrm{Def}(X_n/A_n,B_n)$$

is surjective for all positive integer n and $X_n = F(\alpha_n)(X_{n+1})$. By (2.6.5), the problem becomes to show that

$$H^1(X_{n+1},\mathcal{T}_{X_{n+1}/A_{n+1}}) \longrightarrow H^1(X_n,\mathcal{T}_{X_n/A_n}) \tag{10}$$

is surjective.

Let $f : X_n \to \operatorname{Spec} A_n$ be the deformation of X over A_n, which is projective. By (2.7.1), it follows that

$$H^q(X_n,\Omega^p_{X_n/A_n})$$

is a free A_n-module of rank $h_n^{p,q}$. By cohomology and base change, we have the isomorphisms

$$H^q(X_n,\Omega^p_{X_n/A_n}) \otimes_{A_n} k \xrightarrow{\sim} H^q(X_0,\Omega^p_X),$$

and thus $h_n^{p,q} = h^{p,q}(X)$ is independent on n. In particular,

$$H^0(X,\omega_X) = k \Longrightarrow H^0(X_n,\omega_{X_n/A_n}) = A_n$$

Take a non-trivial section $s : \mathcal{O}_{X_n} \to \omega_{X_n/A_n}$, then $s \otimes_{X_n} X$ is an isomorphism on X. Since ω_{X_n/A_n} is flat over $\operatorname{Spec} A_n$, by (2.6.6), s is an isomorphism. From the perfect pairing

$$\Omega^1_{X_n/A_n} \otimes \Omega^{d-1}_{X_n/A_n} \longrightarrow \omega_{X_n/A_n} \cong \mathcal{O}_{X_n},$$

we have $\mathcal{T}_{X_n/A_n} \cong \Omega^{d-1}_{X_n/A_n}$. Now we may rewrite (10) to

$$A_{n+1}^{h^{d-1,1}} \longrightarrow A_n^{h^{d-1,1}},$$

which is a surjection as desired. □

3 Threefold Calabi-Yau varieties with ordinary double points

We summarize classical results concerning the deformation theory of threefold Calabi-Yau varieties with ordinary double points (ODP). In particular, we state Friedman's criterion for smoothability and the later generalization by Kawamata, Ran and Tian, which shows that deformations are unobstructed once a Kähler resolution exists.

3.1 Setup

Let X_0 be a three-dimensional projective variety with $\omega_{X_0} \cong \mathcal{O}_{X_0}$ and only ordinary double points as singularities. A *small resolution* is a proper birational morphism $f\colon X \to X_0$ such that X is smooth and the exceptional set is a union of rational curves $C_1, \dots, C_r \cong \mathbb{P}^1$ with normal bundle $\mathcal{O}_{C_i}(-1) \oplus \mathcal{O}_{C_i}(-1)$. We denote by $[C_i]$ the homology class in $H_2(X, \mathbb{C})$ or its dual in $H^4(X, \mathbb{C})$. The variety X_0 is called *smoothable* if it occurs as the central fiber of a flat family $X \to \Delta$ with smooth total space and smooth nearby fibers X_t (here Δ is a small disc).

3.2 Friedman's theorem

The first precise criterion for smoothability was given by Friedman [11].

Theorem 3.2.1 (Friedman, 1986)**.** *Let X_0 be as above and let $f\colon X \to X_0$ be a small resolution. Then X_0 is smoothable (and in fact the small resolution can be deformed to an isomorphism) if the following two conditions hold:*

1. *The classes $[C_1], \dots, [C_r]$ generate the cohomology group $H^2(X, \Omega^2_X)$.*
2. *There exists a relation $\sum_{i=1}^r a_i[C_i] = 0$ in $H^4(X, \mathbb{C})$ with all coefficients $a_i \neq 0$.*

3.3 Unobstructedness and the theorem of Kawamata-Ran-Tian

Subsequently it was shown independently by Y. Kawamata, Z. Ran and G. Tian [12, 13, 14] that the first generating condition is superfluous provided that a Kähler resolution exists. The crucial point is a generalization of the Bogomolov-Tian-Todorov unobstructedness theorem to singular Calabi-Yau varieties.

Theorem 3.3.1 (Kawamata, Ran, Tian)**.** *Let X_0 be a threefold with $\omega_{X_0} \cong \mathcal{O}_{X_0}$ and only ordinary double points. Assume there exists a resolution $p\colon X' \to X_0$ such that X' is a compact Kähler manifold. Then the versal deformation space* $\mathrm{Def}(X_0)$ *is smooth, i.e. deformations of X_0 are unobstructed. In particular, the condition that the exceptional curves generate $H^2(\Omega^2_X)$ in Friedman's theorem can be dropped.*

3.4 Sketch of proofs

3.4.1 Friedman's smoothing criterion

1. **Deformations of a small resolution.** Let $f\colon X \to X_0$ be a small resolution with exceptional curves C_i. A deformation of X simultaneously blows down the C_i if the associated infinitesimal deformation maps the normal bundle sequence so that the curves are obstructed from deforming. More precisely, the local-to-global exact sequence for Ext^1 and Ext^2 relates the first-order deformations of X and of the formal completion along each C_i. Condition *(a)* (generating $H^2(\Omega_X^2)$) guarantees that the natural map $\mathrm{Ext}^1(\Omega_X^1, \mathcal{O}_X) \to \bigoplus_i \mathrm{Ext}^1(\mathcal{O}_{C_i}, \mathcal{O}_{C_i}(-1)^{\oplus 2})$ is surjective, so one can lift local smoothings of the double points to a global deformation.

2. **Simultaneous contraction and the relation among $[C_i]$.** In order to contract all C_i simultaneously in a family, the exceptional set must become trivial in homology. The obstruction to contracting a curve in a deformation is its fundamental class; a family of threefolds containing X as a fiber and for which the curves disappear corresponds to a first-order deformation $\nu \in H^1(X, \Theta_X)$ for which the restriction to each C_i vanishes. This translates precisely to the existence of a relation $\sum a_i[C_i] = 0$ in $H^4(X, \mathbb{C})$ with every $a_i \neq 0$. The coefficients a_i determine the "speed" at which each singularity is smoothed.

3. **Conclusion.** Once one has such a first-order deformation, the unobstructedness (proved in 3.4.2) ensures that it can be lifted to an actual smoothing of X_0. The work of Kawamata-Ran-Tian shows that the extra generating condition is automatically satisfied when a Kähler resolution exists, completing the picture.

3.4.2 Unobstructedness for ODP Calabi-Yau threefolds

1. **Deformation functor and tangent-obstruction theory.** Since X_0 is a local complete intersection (ODP), its deformation theory is controlled by the cotangent complex $\mathbb{L}_{X_0}$. The tangent space is $\mathrm{Ext}^1(\Omega_{X_0}^1, \mathcal{O}_{X_0})$ and obstructions lie in $\mathrm{Ext}^2(\Omega_{X_0}^1, \mathcal{O}_{X_0})$. Smoothness of $\mathrm{Def}(X_0)$ follows if $\mathrm{Ext}^2(\Omega_{X_0}^1, \mathcal{O}_{X_0}) = 0$.

2. **Comparison with a Kähler resolution.** Let $p\colon X' \to X_0$ be a Kähler resolution (not necessarily small). The sheaf of differentials $\Omega_{X_0}^1$ sits in an exact triangle relating it to $p_*\Omega_{X'}^1$ and a sheaf supported on the singular points. Using local calculations one shows the connectedness of the deformation functors, namely $\mathrm{Def}(X_0) \cong \mathrm{Def}(X')$ near the central fiber. Hence it is enough to prove $\mathrm{Ext}^2(\Omega_{X'}^1, \mathcal{O}_{X'}) = 0$ for the smooth Kähler threefold X'.

3. **Hodge theory and the Bogomolov–Tian–Todorov argument.** On the smooth compact Kähler manifold X' we have the Hodge decomposition and Serre duality. Because p contracts only rational

curves, one still has $\omega_{X'} \cong \mathcal{O}_{X'}$. The classical unobstructedness proof for Calabi–Yau manifolds (due to Bogomolov, Tian, Todorov) uses the contraction map

$$H^2(X', \Theta_{X'}) \times H^1(X', \Theta_{X'}) \to H^3(X', \Theta_{X'})$$

given by the cup product with the trace map $\Theta_{X'} \otimes \Omega^p_{X'} \to \Omega^{p-1}_{X'}$. Under the Calabi–Yau condition $\omega_{X'} \cong \mathcal{O}_{X'}$, Serre duality identifies the obstruction space $\mathrm{Ext}^2(\Omega^1_{X'}, \mathcal{O}_{X'}) \cong H^2(X', \Theta_{X'})$ with the dual of $H^1(X', \Omega^2_{X'})$. By a clever use of the local Torelli theorem for the period map, one shows that the multiplication map vanishes, hence obstructions vanish. This argument adapts to the present situation and yields $\mathrm{Ext}^2(\Omega^1_{X_0}, \mathcal{O}_{X_0}) = 0$.

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